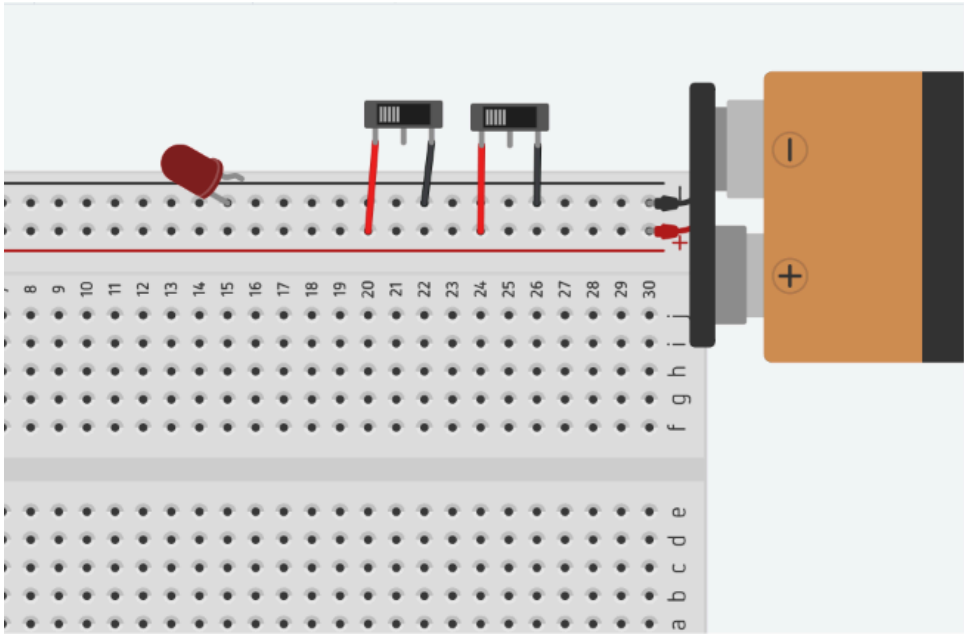
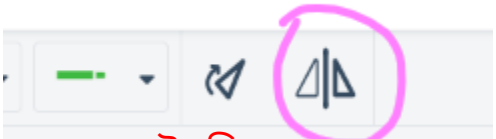


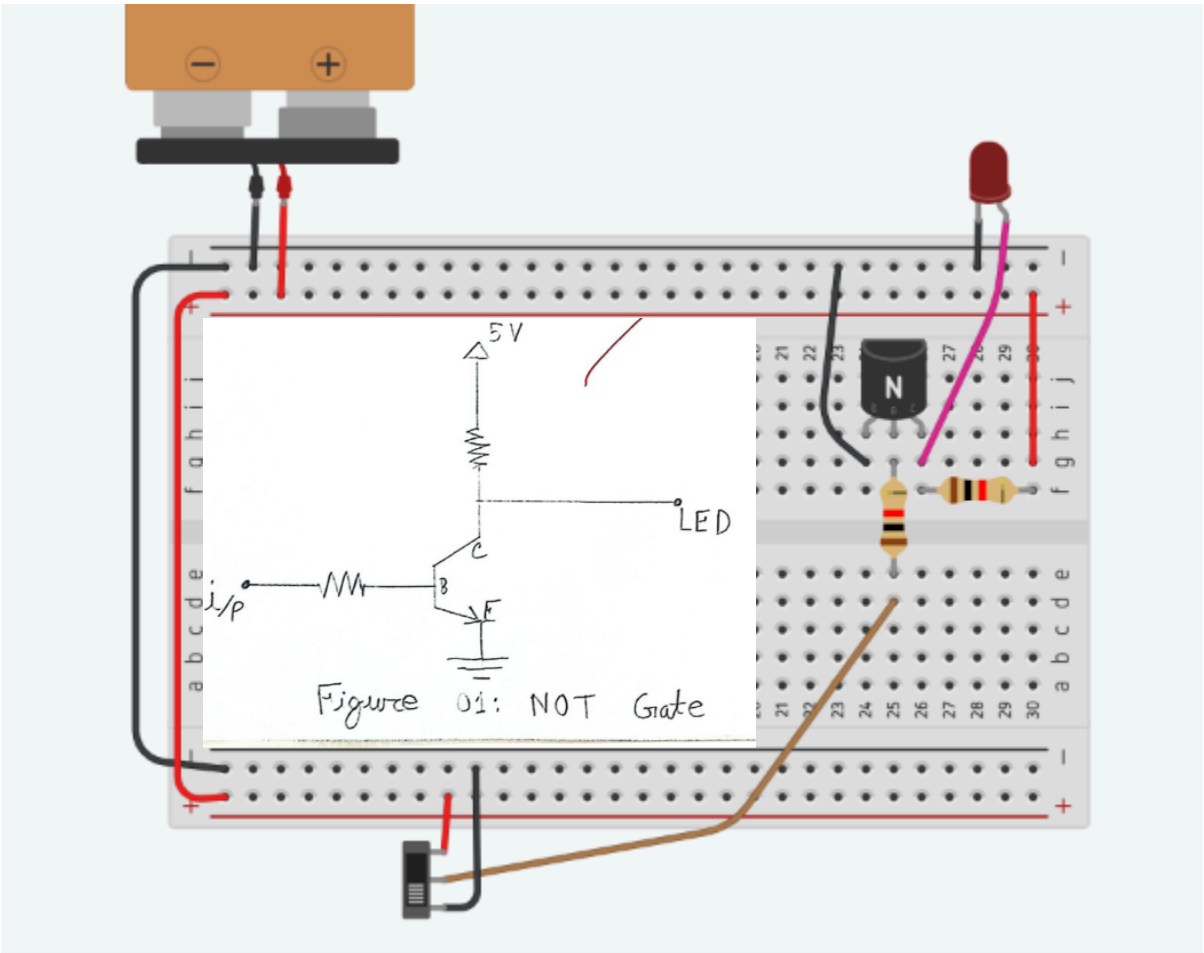
এটা initial setup rtl, dtl, ttl, nmos, cmos এর জন্য। ল্যাবে trainer board থাকে। তার ভিতরেই এই setup আছে।

Battary er positive = +5v
Battary er negative = ground
Switch = switch
Led = Led



TINKER CAD এ একটা **mirror** বাটন আছে যেটা দিয়া **transistor** গুলা কে ঘুরিয়ে নিয়েছি। কারণ **lab** এ **serial e-b-c** তে থাকে।

RTL NOT GATE



https://www.tinkercad.com/things/4P04vV9d50D/editel?returnTo=%2Fdashboard&sharecode=g1FPPuhiqH-DBb8HMoO6mqSmx1e4FtOsfNLwEscxHOE

RTL NAND GATE

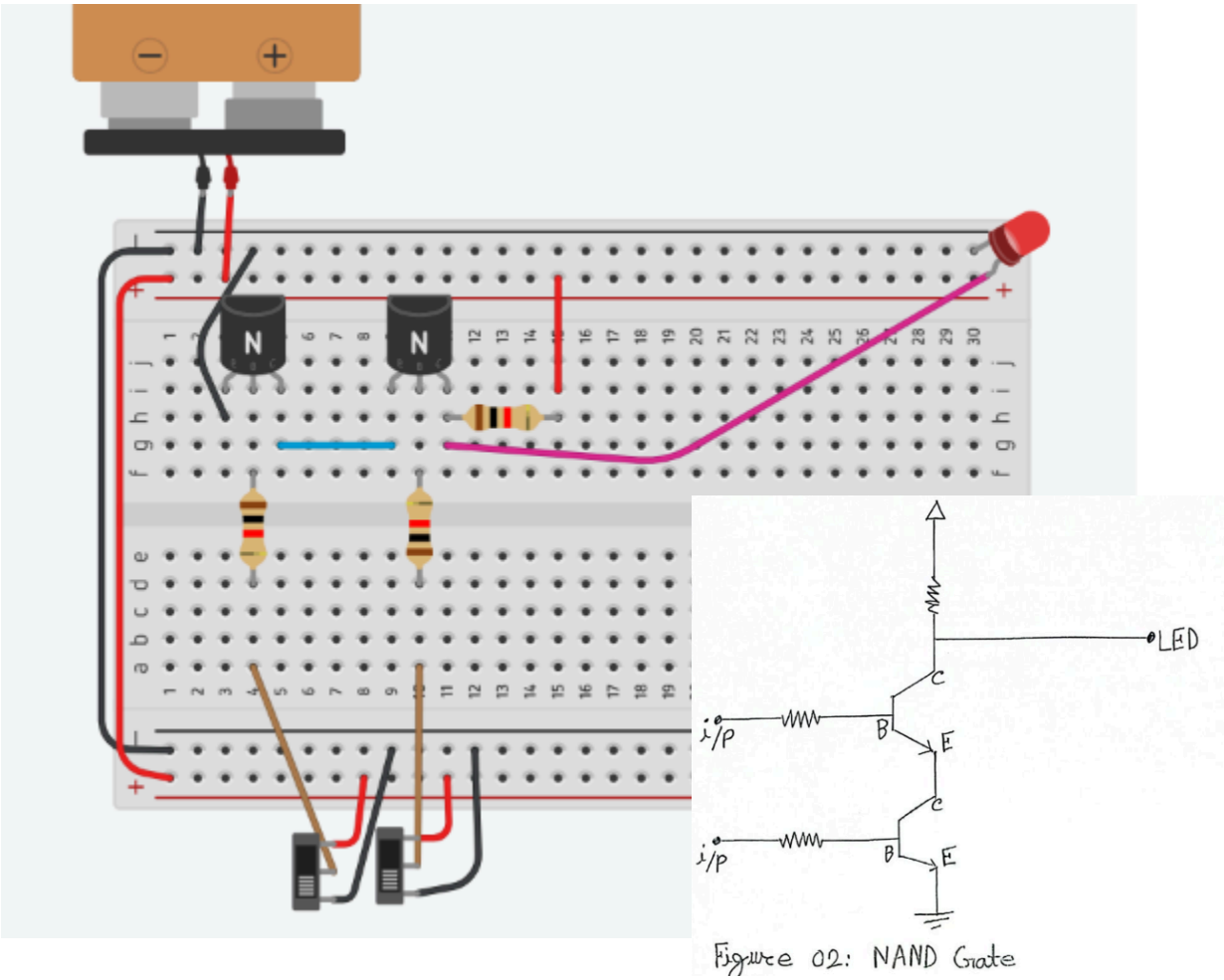
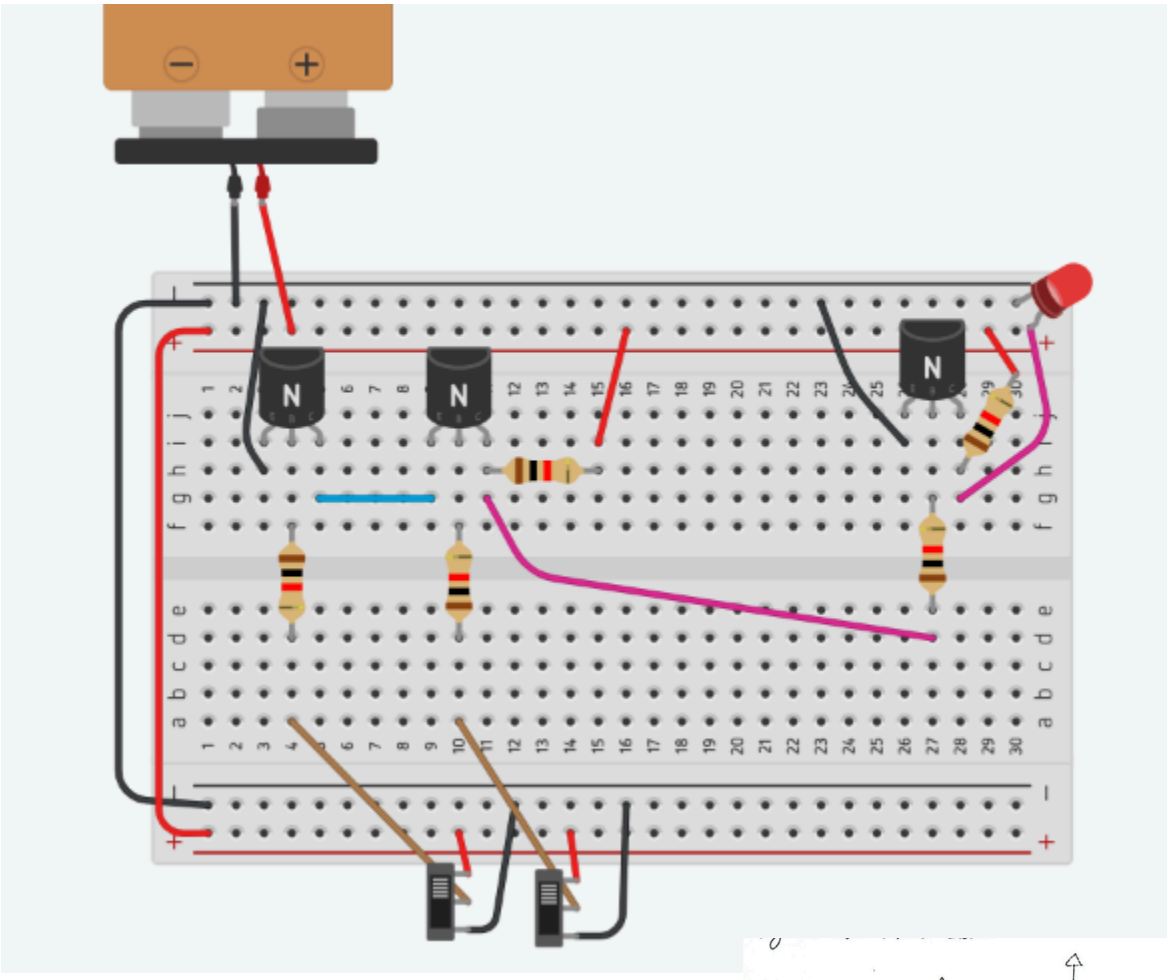


Figure 02: NAND Gate

https://www.tinkercad.com/things/h5YYrnXQrns/editel?returnTo=%2Fdashboard&sharecode=gDYERmwZECJVooD89k_GPcUWrK5dA34cBXtRhpEBrPA

RTL AND GATE



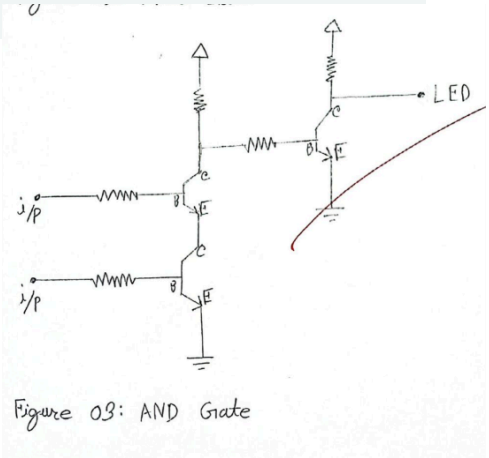


Figure 03: AND Gate

https://www.tinkercad.com/things/3xZYKZLCvP3/editel?returnTo=%2Fdashboard&sharecode=pKnSq5zqazvGmkCcZnKcB-euhP74JeJsf6s5dxoHJI0

RTL NOR GATE

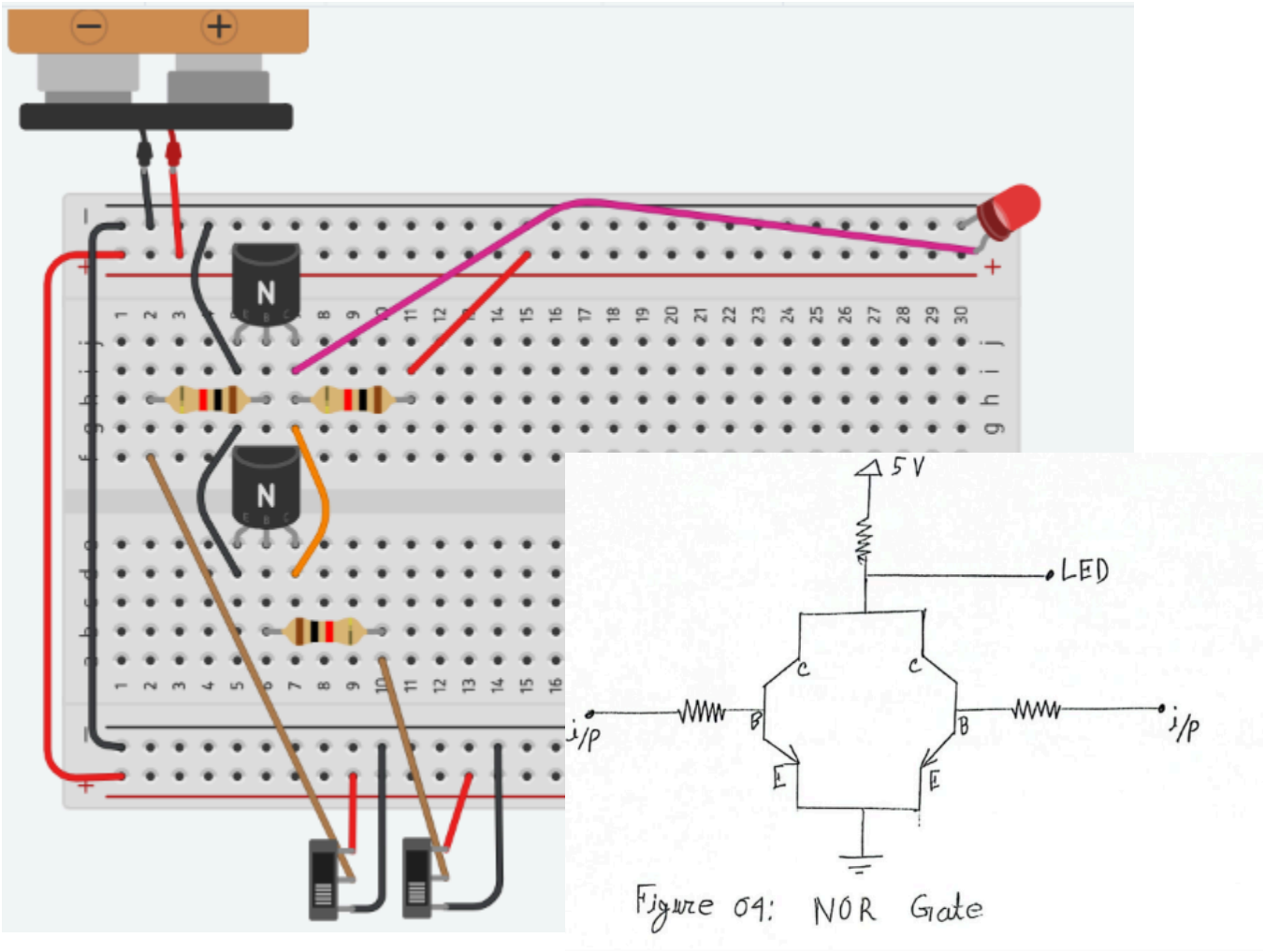


Figure 04: NOR Gate

https://www.tinkercad.com/things/clzTsMLrTid/editel?returnTo=%2Fdashboard&sharecode=Y4kEeH64P1hEAtruXEUOiL5qnBL-LBV85y-EbQIkdW8

RTL OR GATE

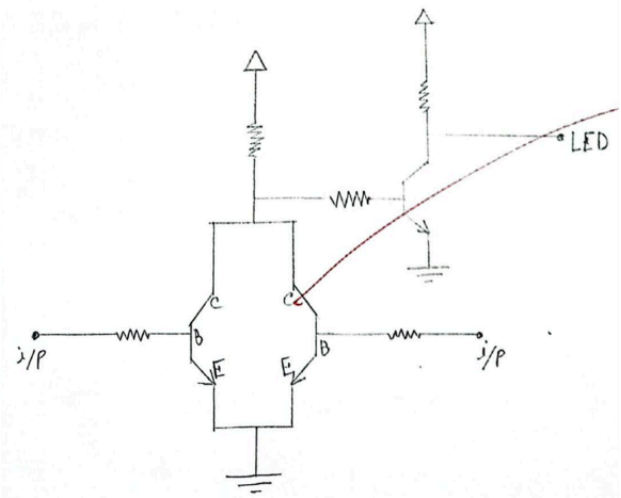
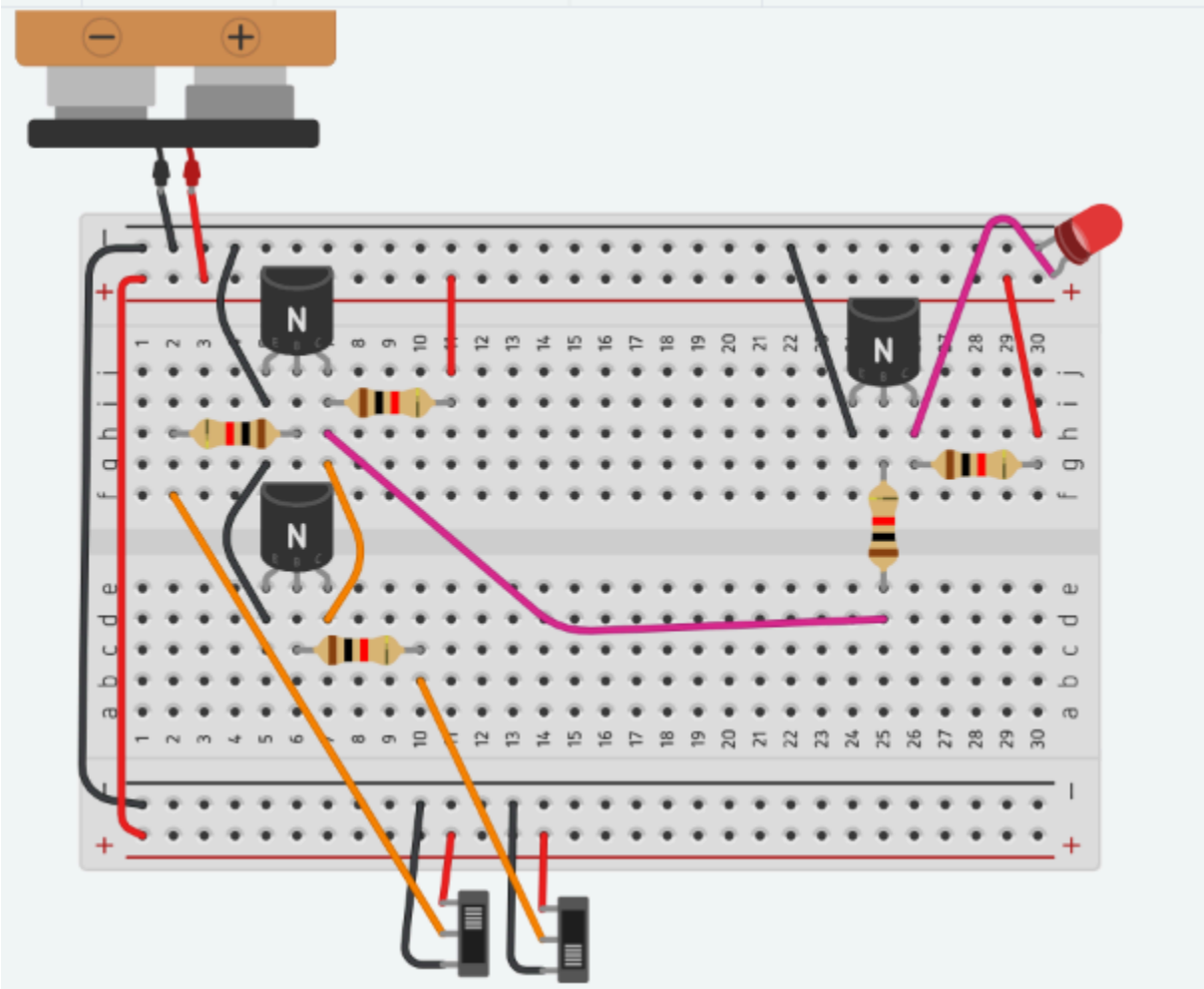
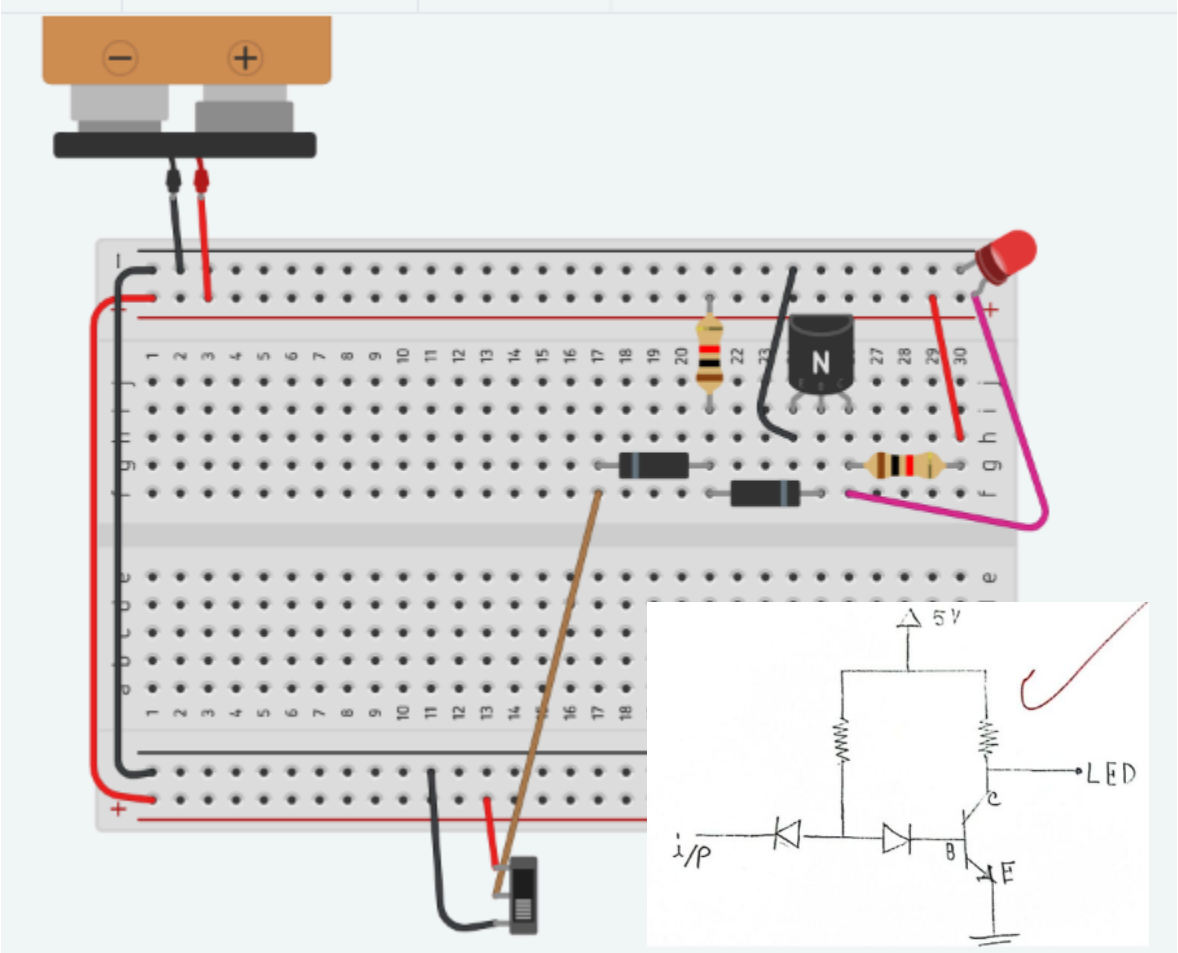


Figure 05: OR Gate



https://www.tinkercad.com/things/0jNX2IgTxXO/editel?returnTo=%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=j5JXJWcDz1YWx9sOYd_W895Z8IqhCUxPIV1cQz3nSoM

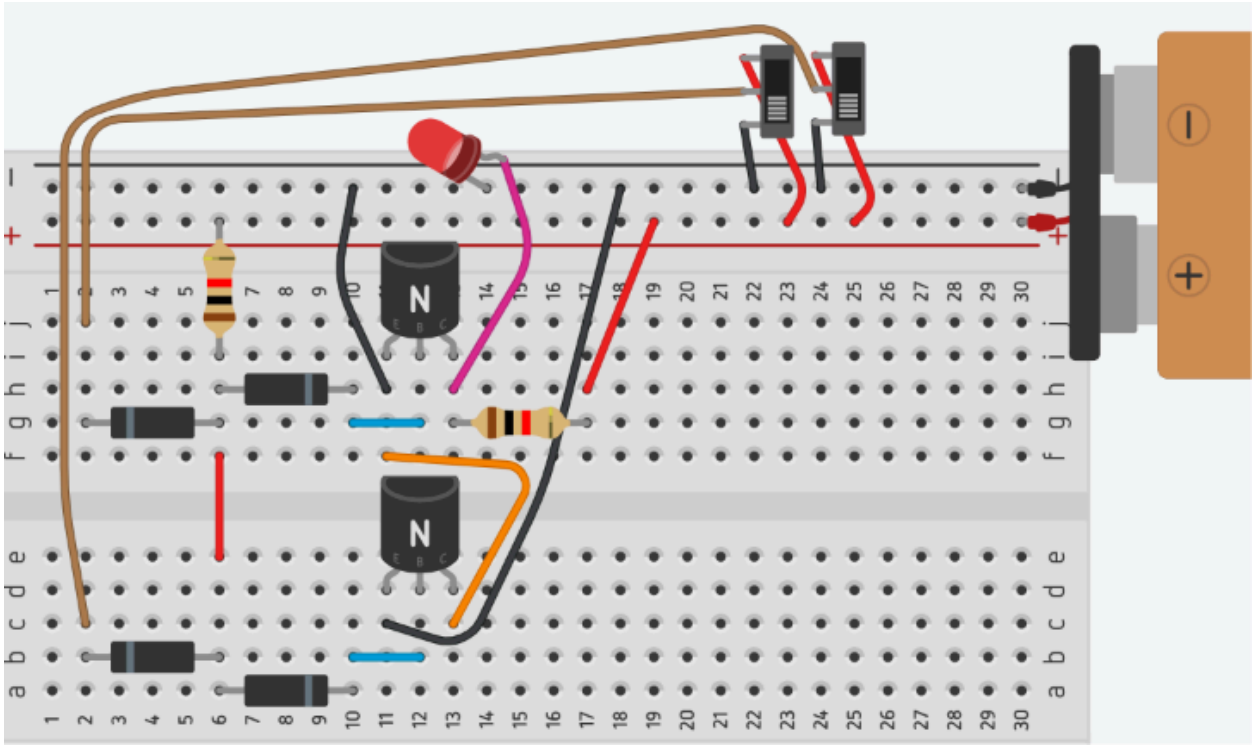
DTL NOT GATE

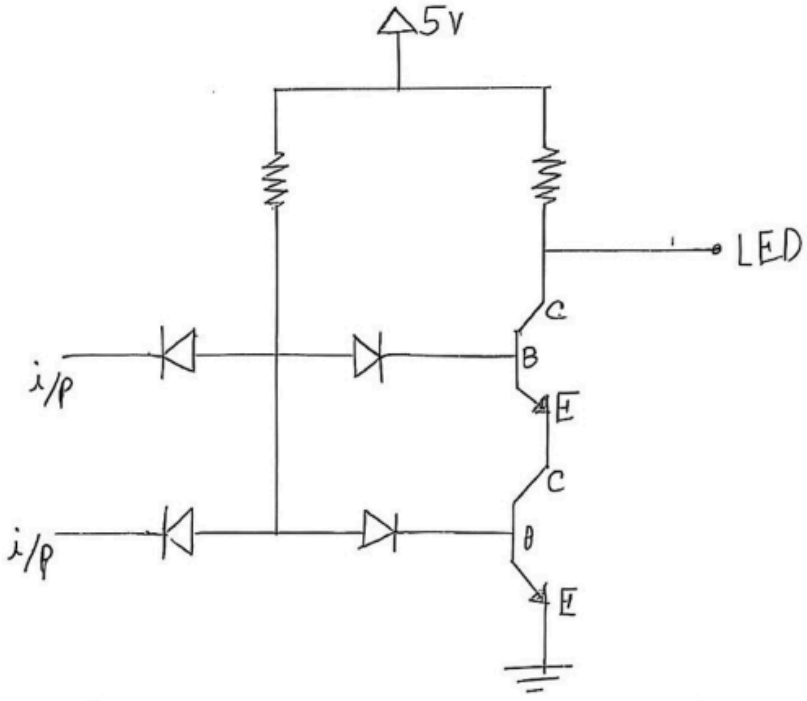


https://www.tinkercad.com/things/70VGGqtehqR-dtl-not-gate/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=twhpoLIPfRV-D8ipWWChDp9gFGTHsrgTYUYlCJ6b0AM

DTL NAND GATE

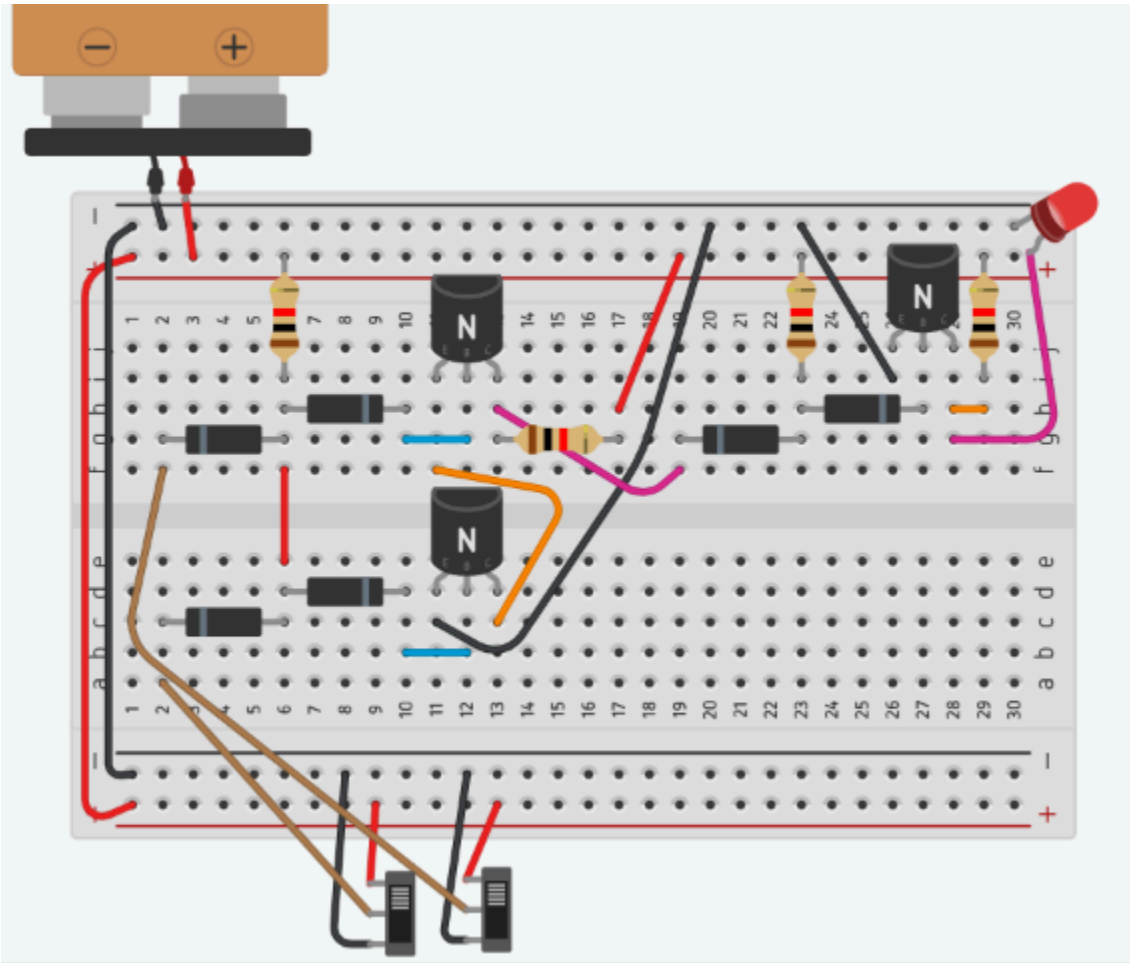
https://www.tinkercad.com/things/6JRLcWbEsup-dtl-nand-gate/editel?returnTo=%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=G0LtzebHNwtxVjPzaouucPt4XjGS9ZTyEnBGlU-if9A

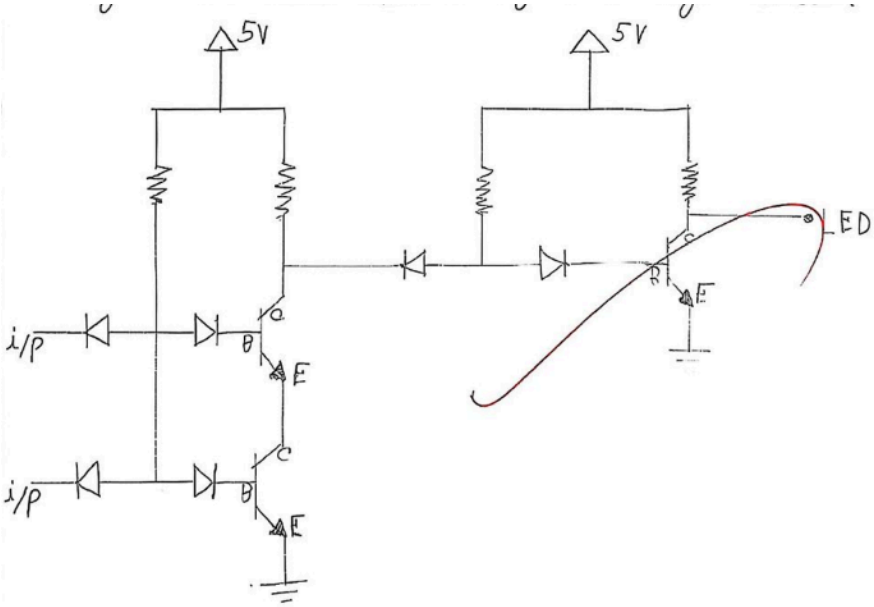




DTL AND GATE

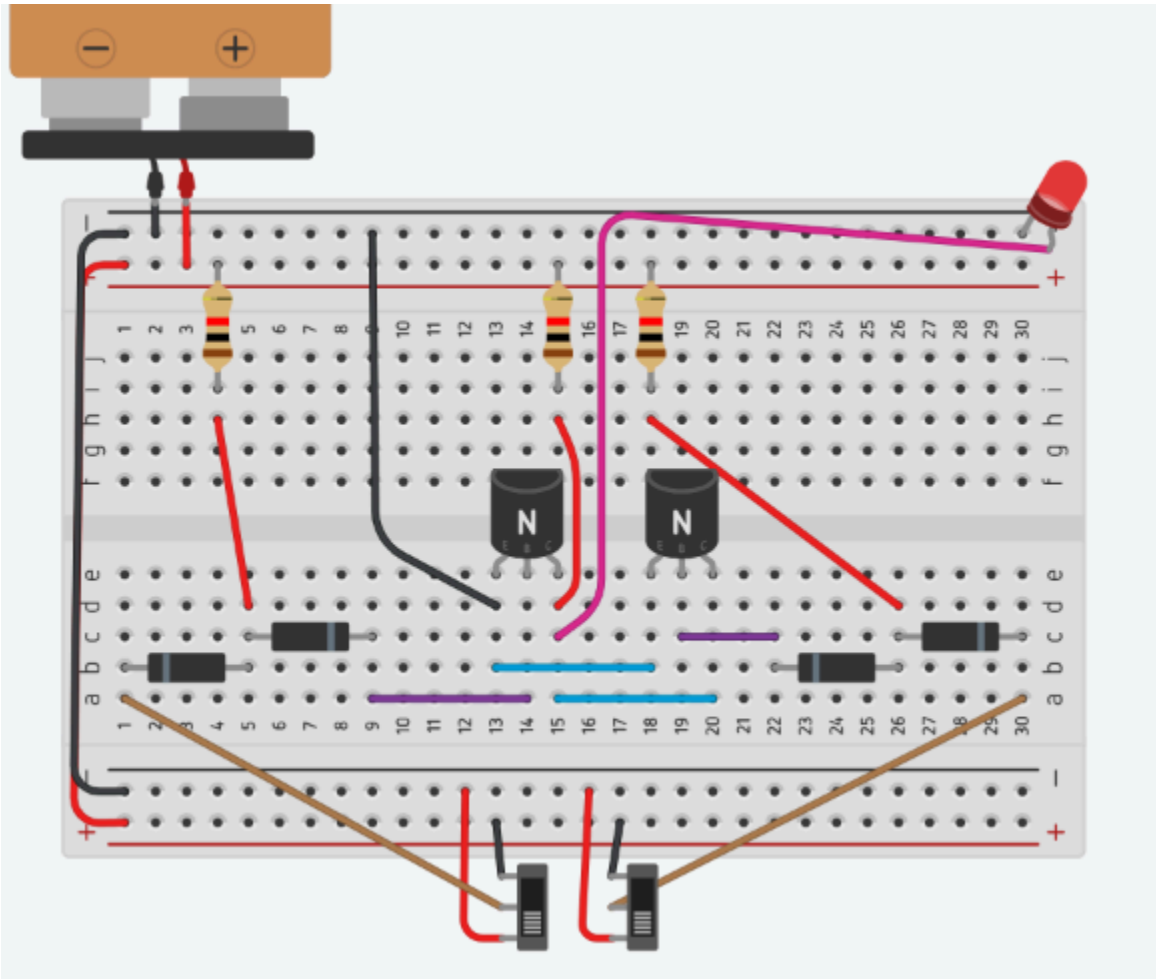
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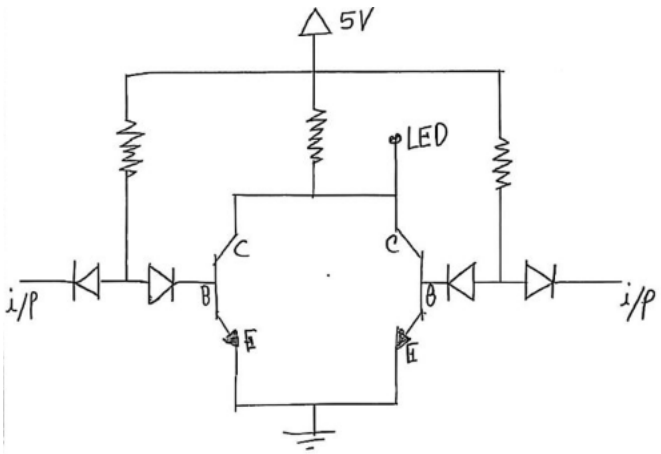


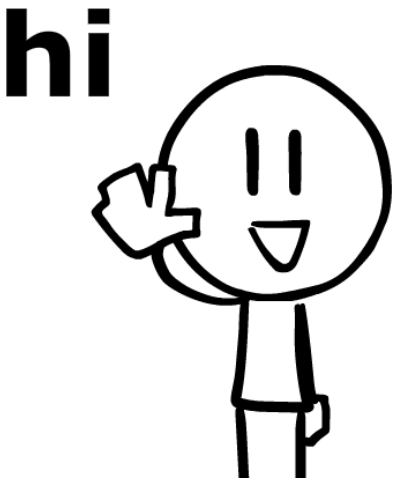


DTL NOR GATE

https://www.tinkercad.com/things/7rUSfdaTsuP/editel?returnTo=%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=2kaRfOst1oD5G5ydqjIMyLbVltnVPEGmKeIyPqRuAGQ

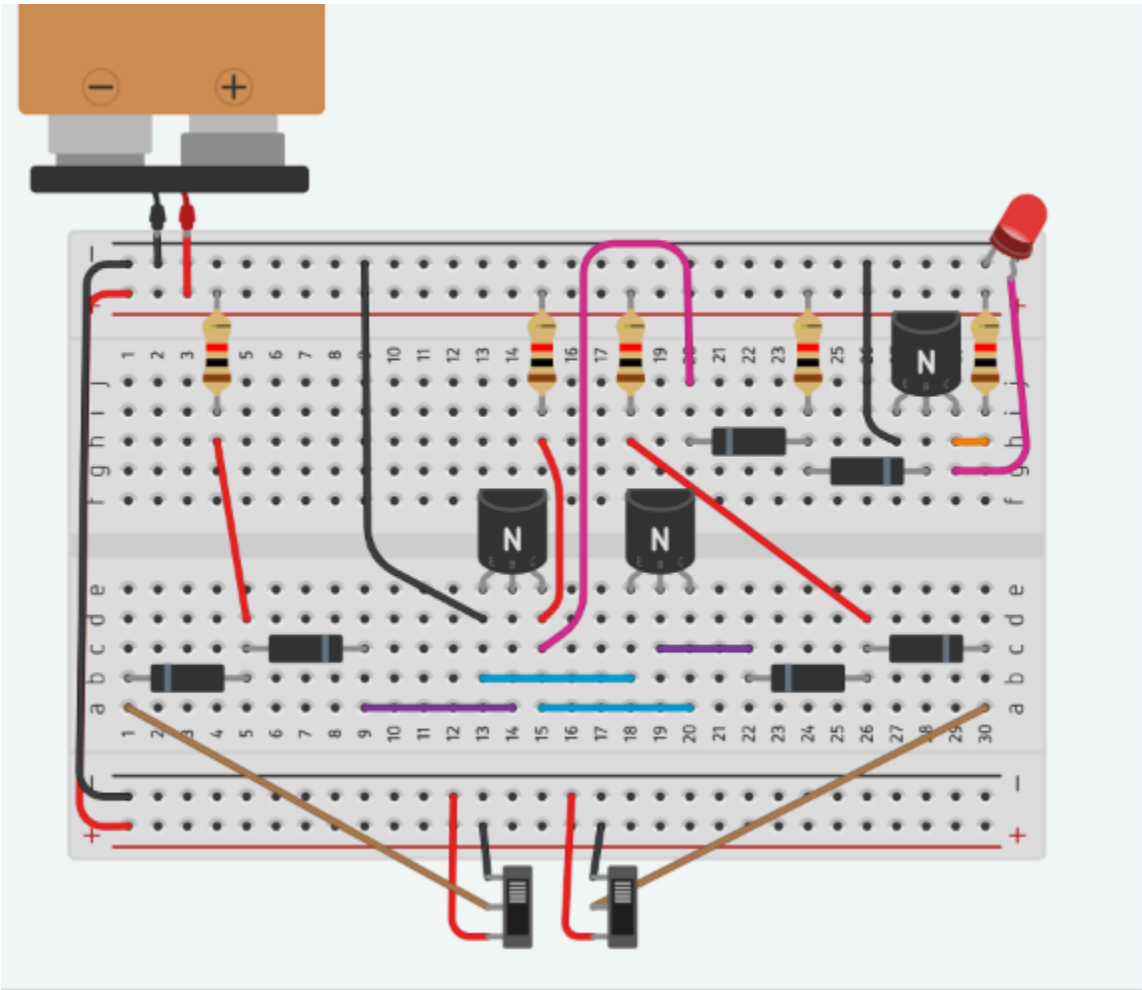






DTL OR GATE

https://www.tinkercad.com/things/44BPReR7eaI/editel?returnTo=%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=4BoPbia9lfthrLj9MQ4j7RLcY5kc4dRvQA9V_AXX1co



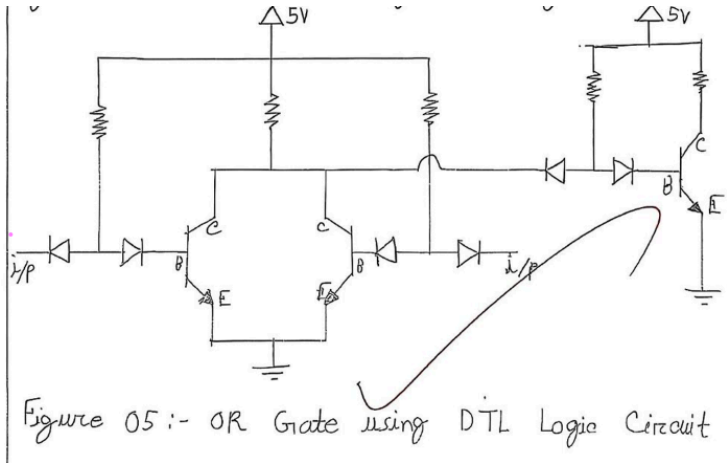
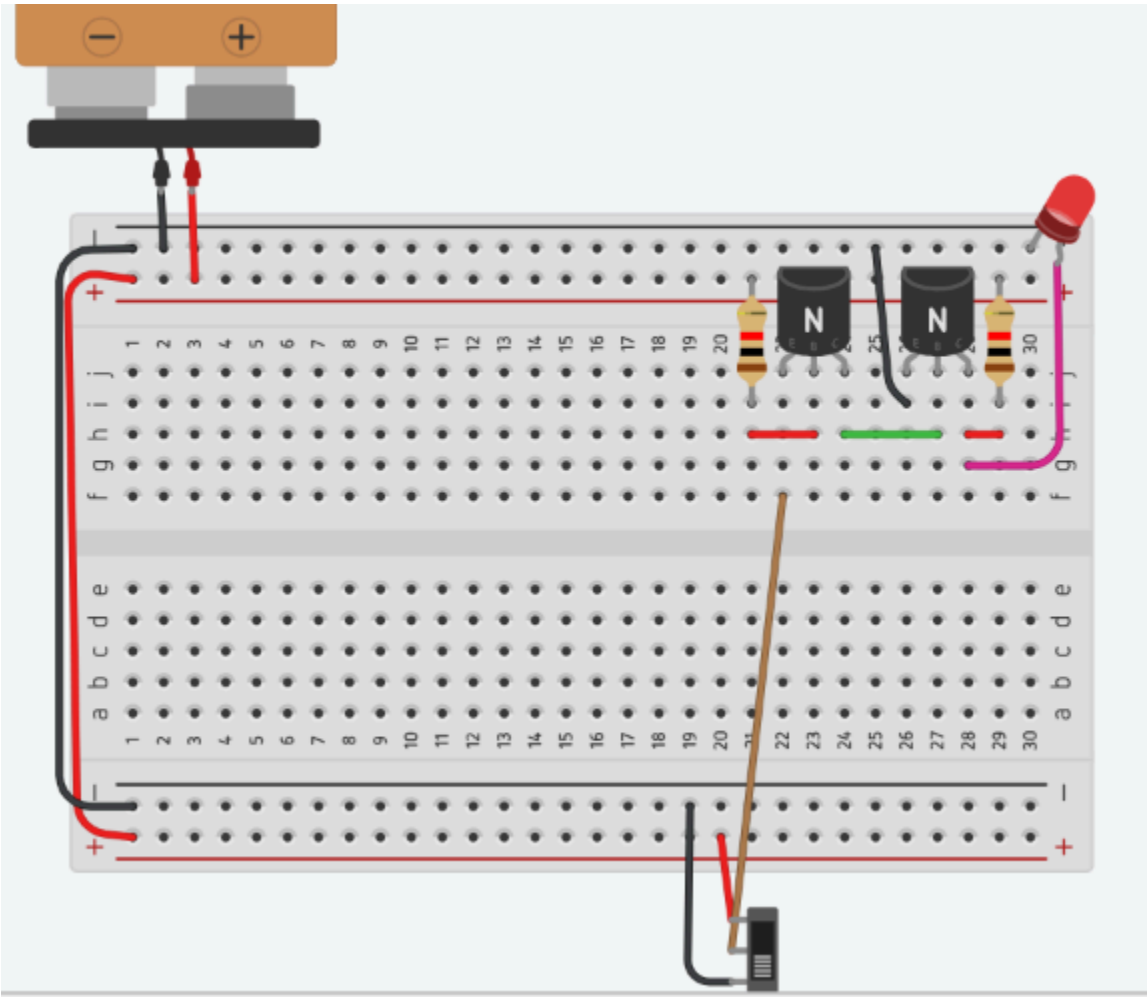


Figure 05 :- OR Gate using DTL Logic Circuit

TTL NOT GATE

(LED পুইরা যাইতে ছিল তাই একটা resistance add করছি LED এর সাথে

)https://www.tinkercad.com/things/eO7rq9ms5k1/editel?returnTo=%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=sragTwFWN2hbUpJLpbMRAQWZ7-T9omkkYdWmuv6sJdQ



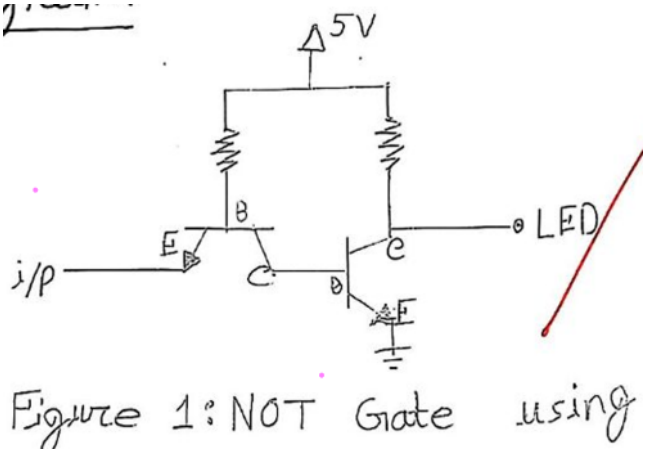
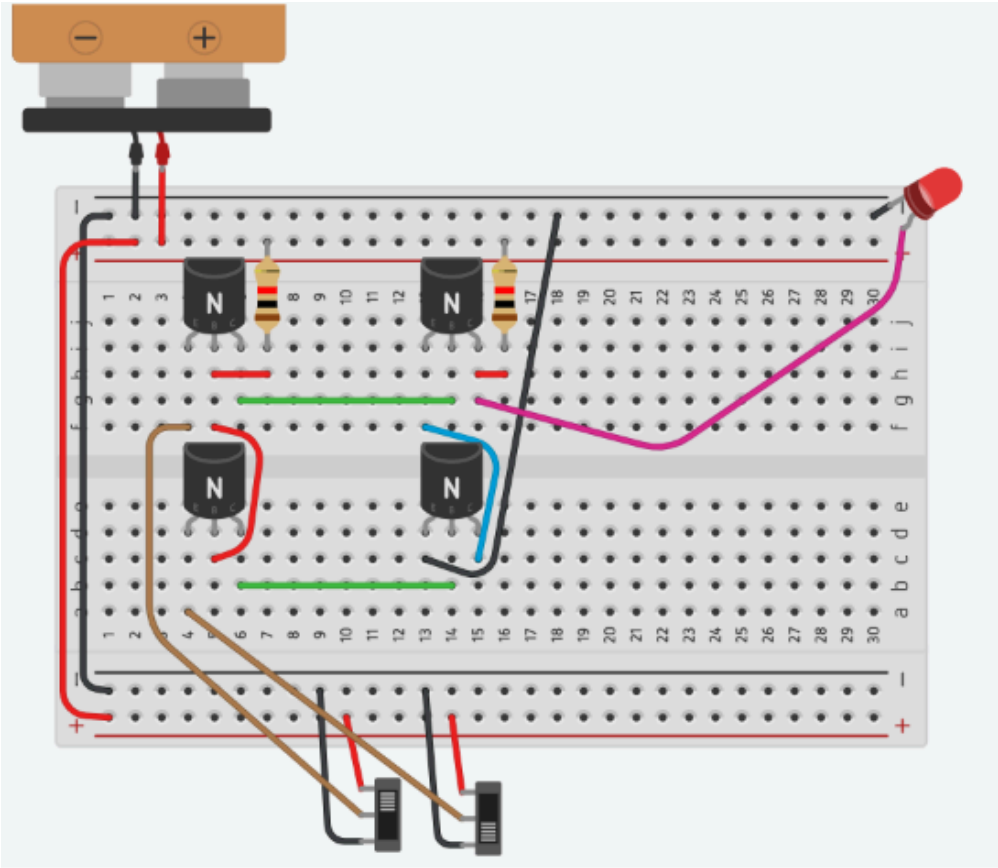
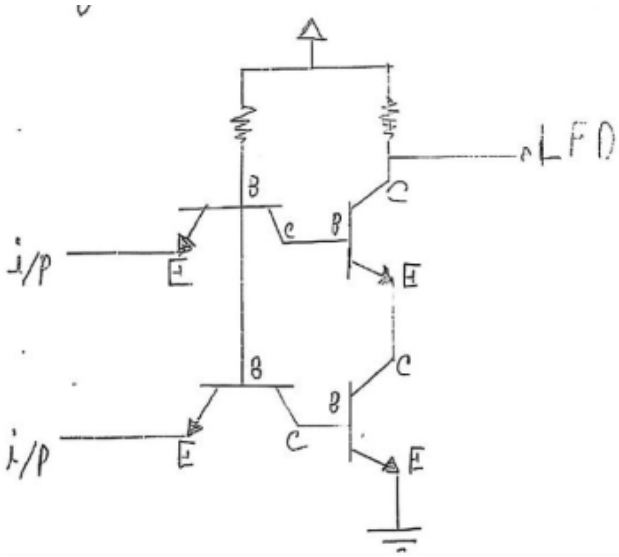


Figure 1: NOT Gate using

TTL NAND GATE

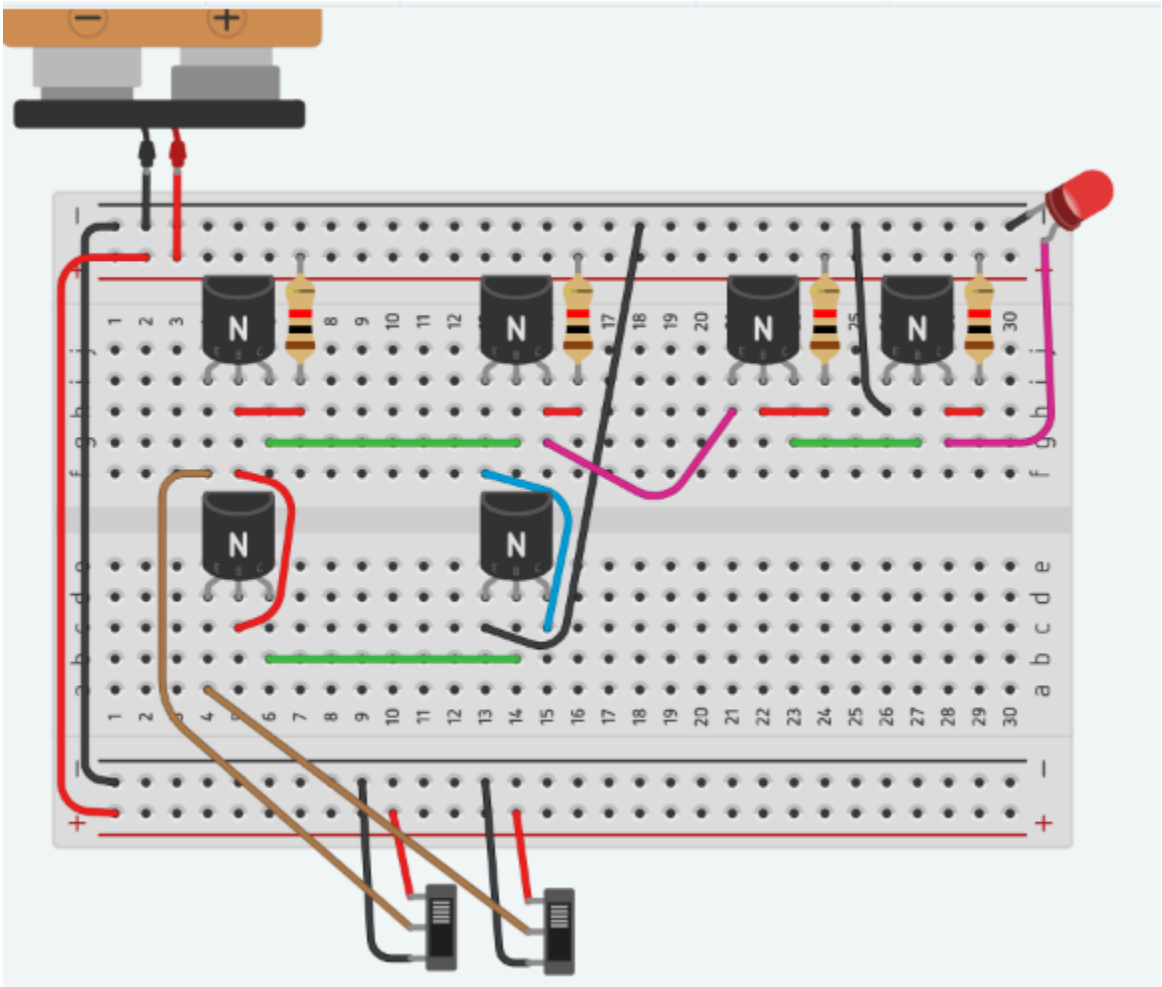
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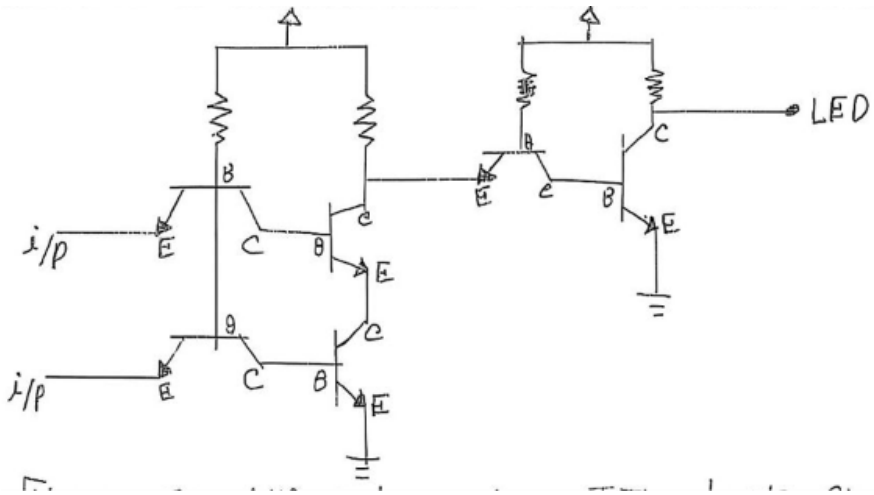




TTL AND GATE

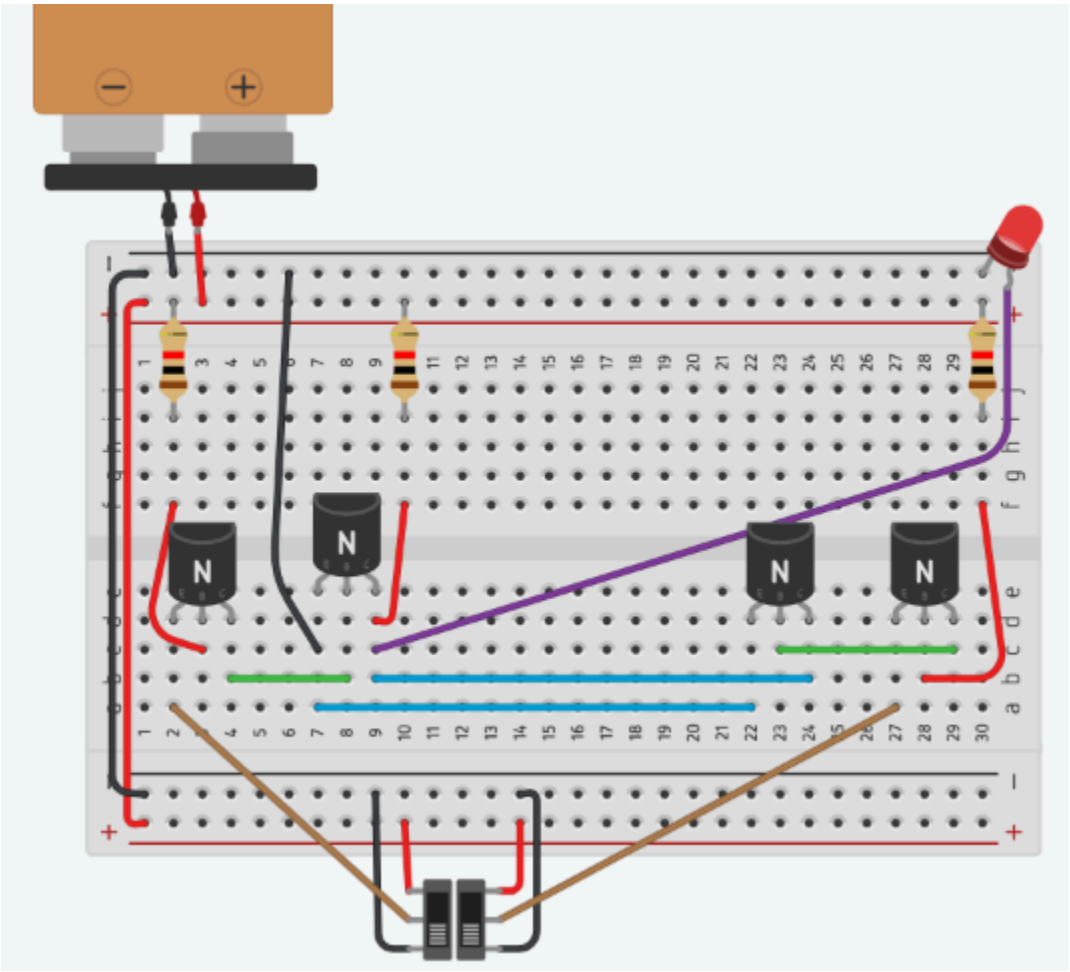
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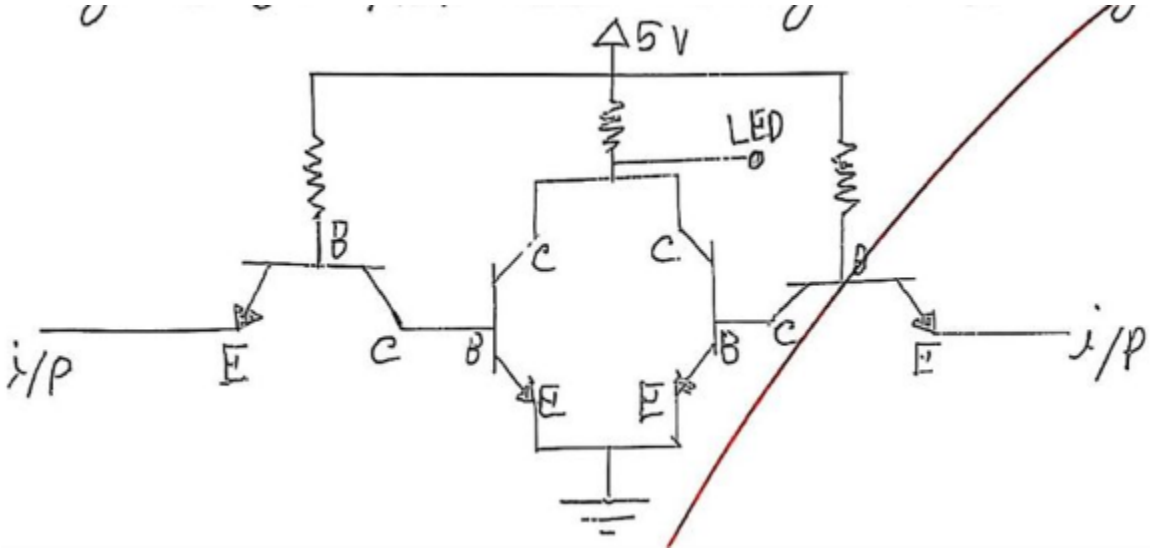




TTL NOR GATE

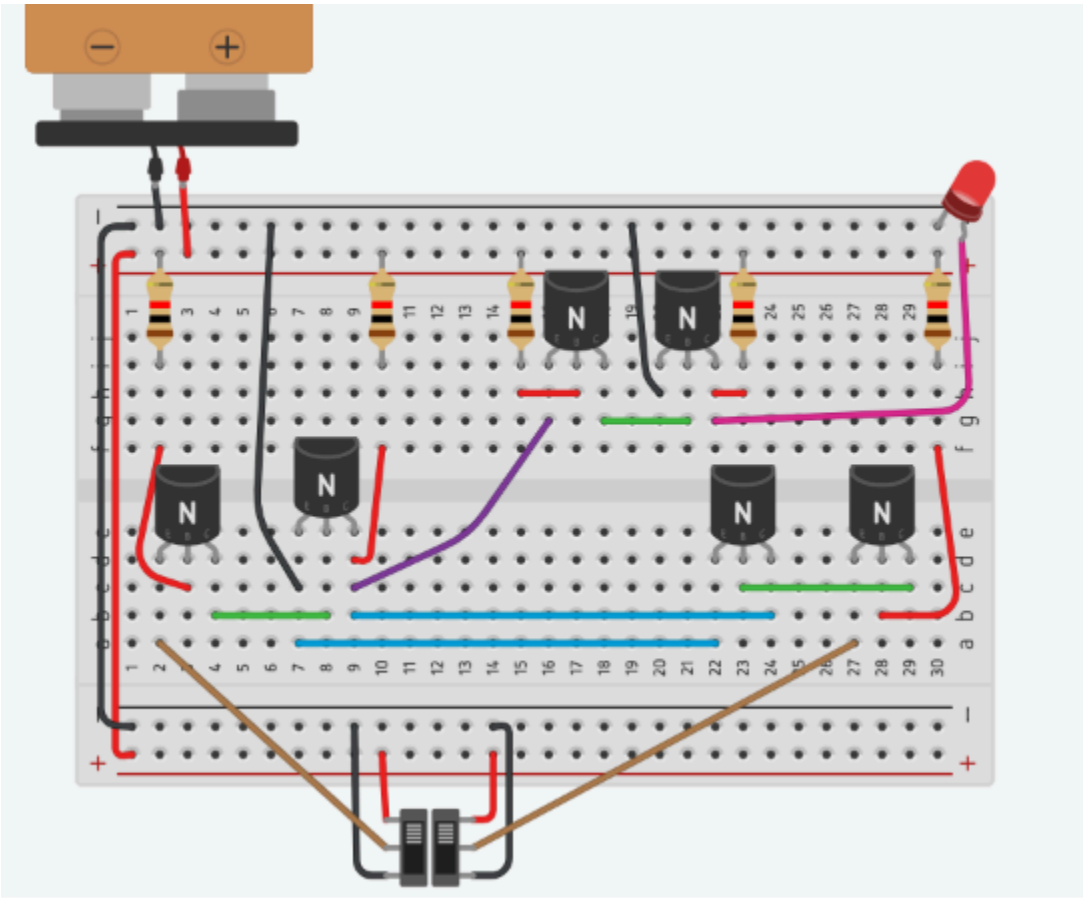
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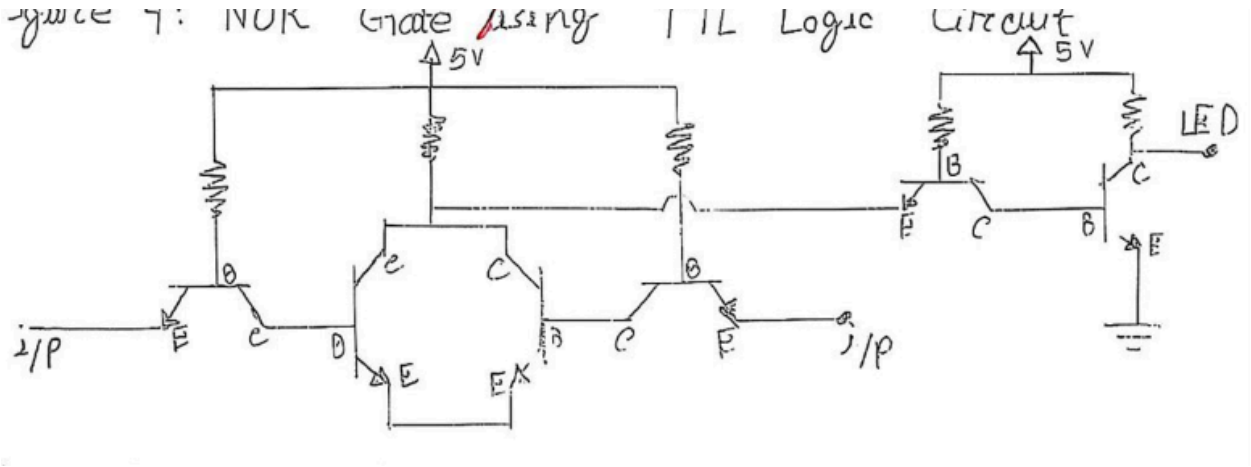




TTL OR GATE

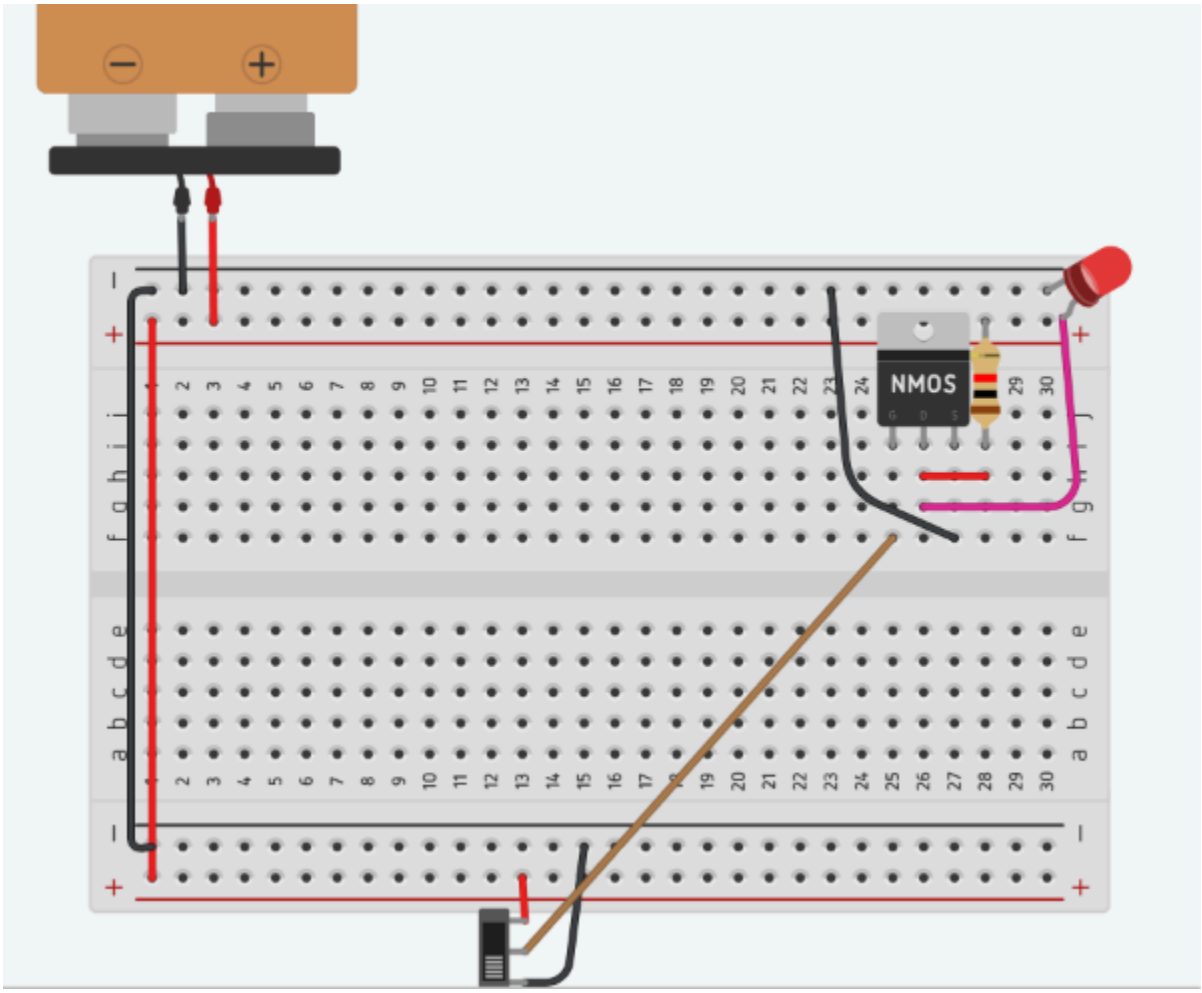
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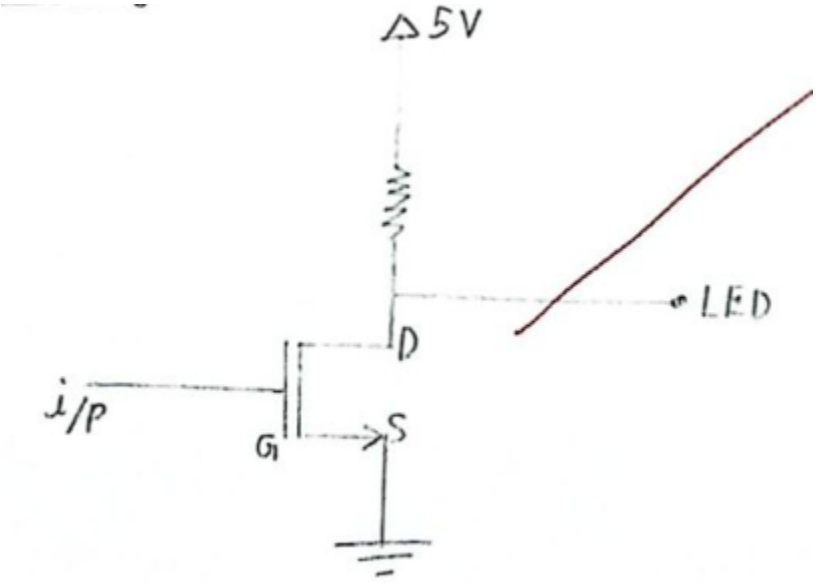




NMOS NOT GATE

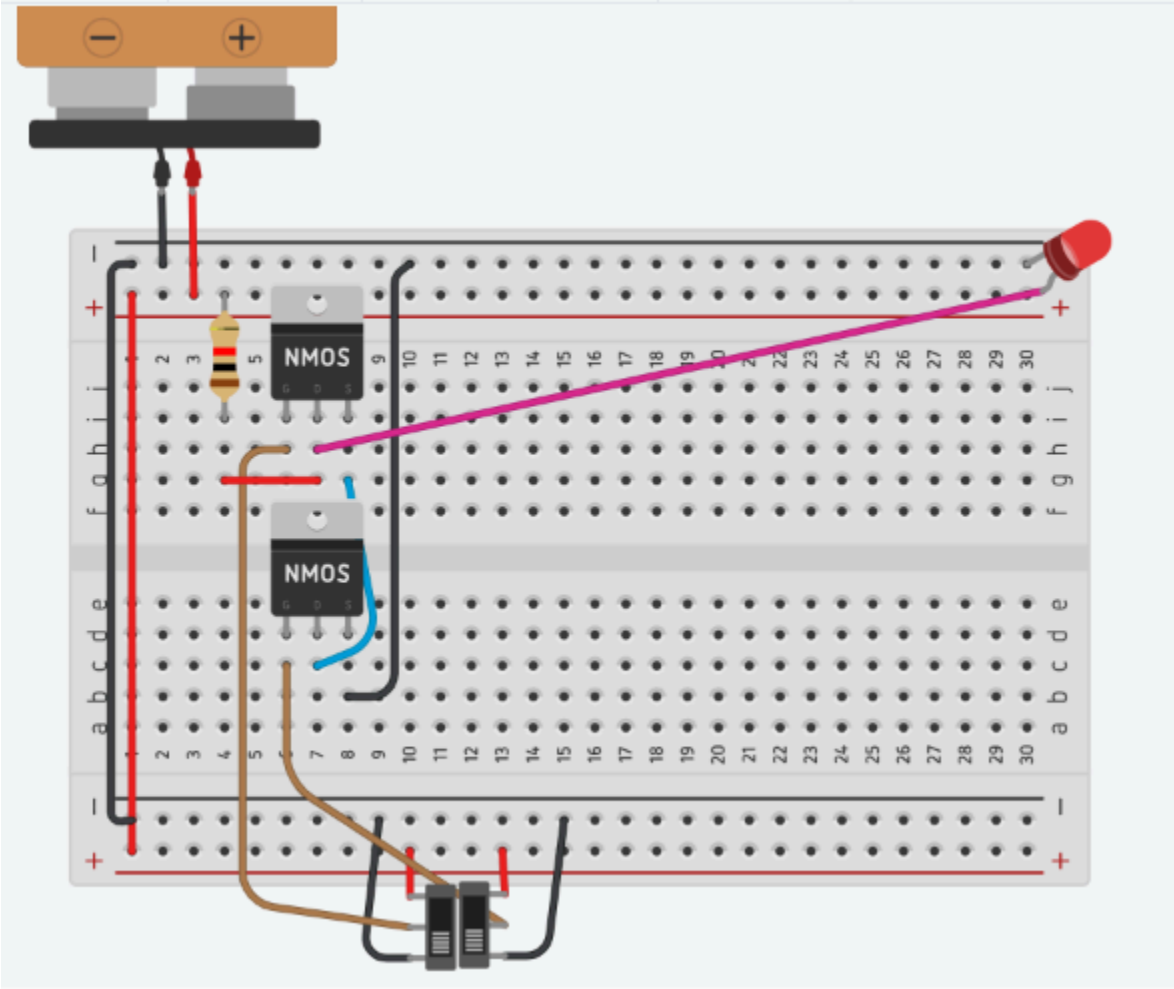
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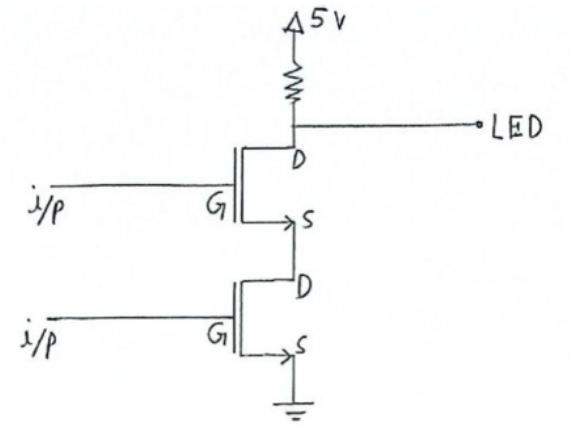




NMOS NAND GATE

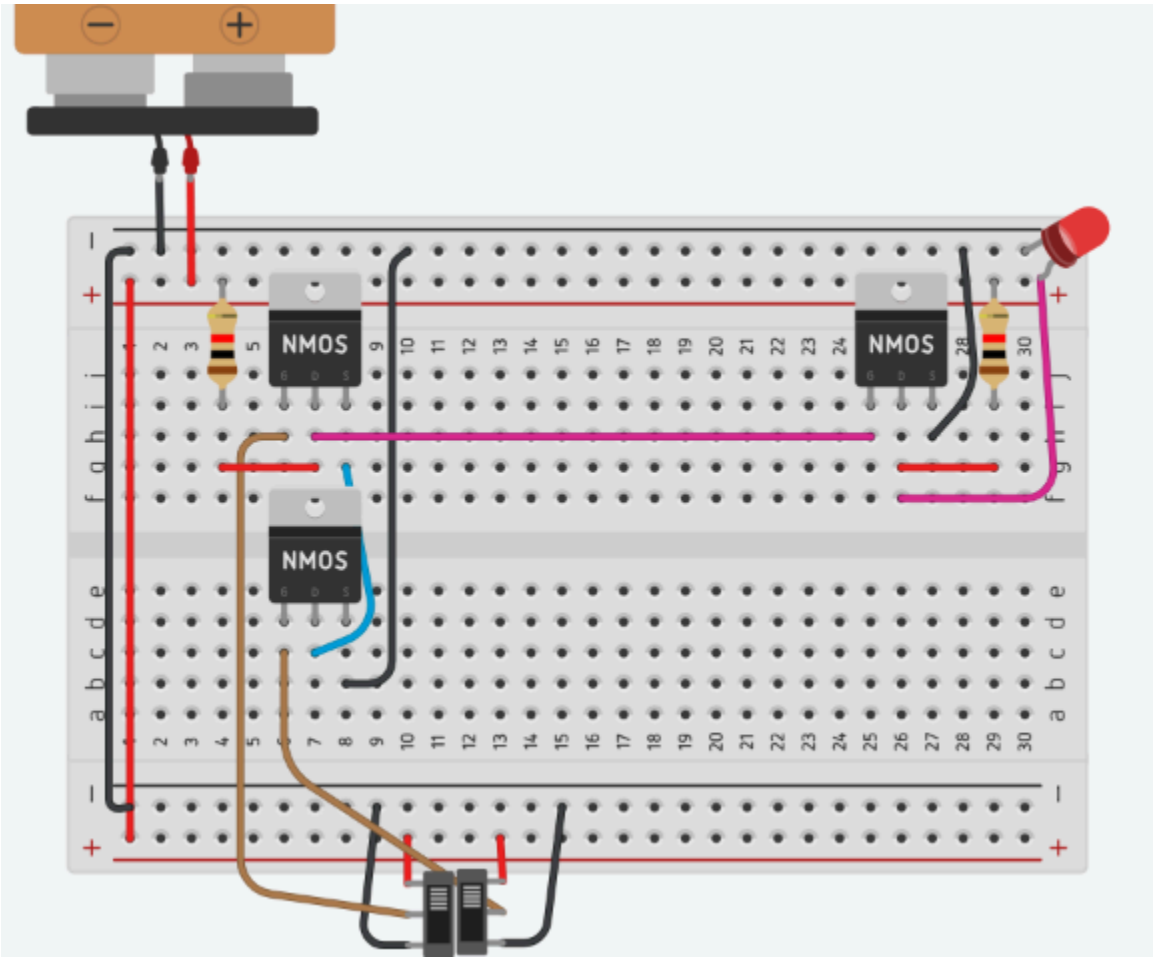
https://www.tinkercad.com/things/j17OgCWL6vH/editel?returnTo=%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=RkJLtqbnROichi685BIFeiPj6vwbhzGR5s9rNRz1Oec

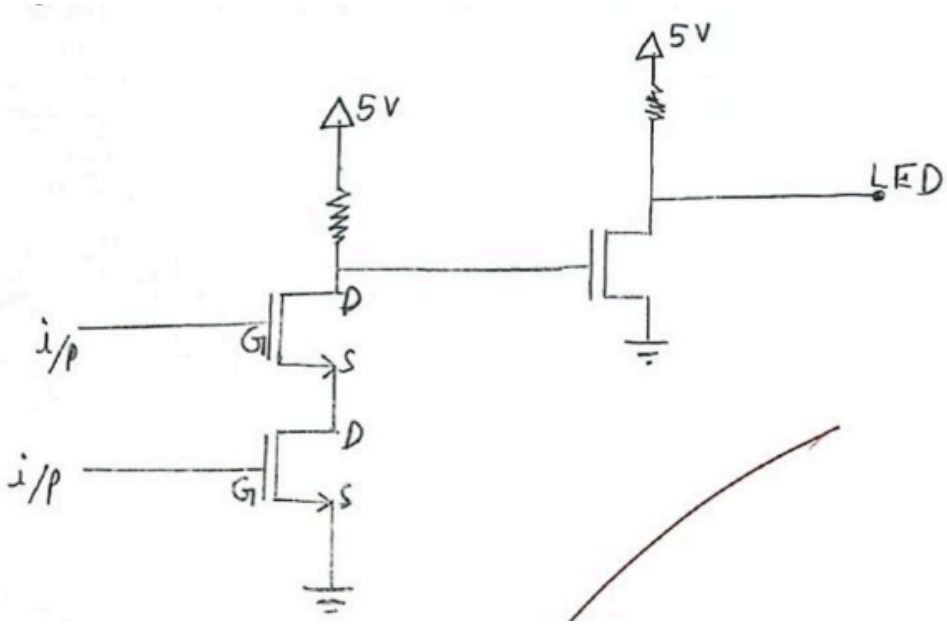




NMOS AND GATE

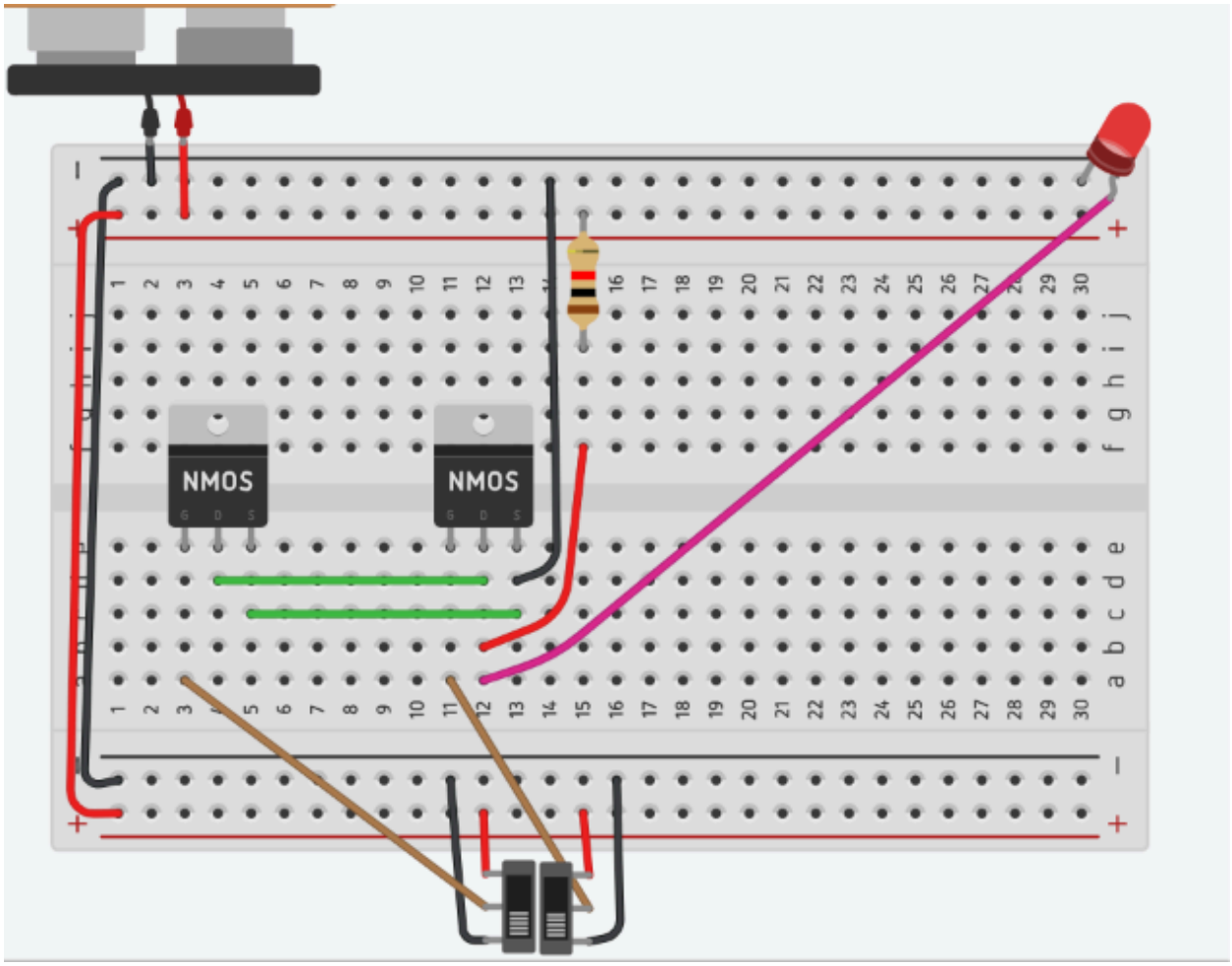
https://www.tinkercad.com/things/7AhtqbF68sk/editel?returnTo=%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=0gU8Y-eZUImzQn8antkik9TGCO-h5QYUTTYxZkq6Jyw





NMOS NOR GATE

https://www.tinkercad.com/things/hhG9ureMwf0/editel?returnTo=%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=pegN4I7Bp1NoWbPXg0cK3W13AaEPc_RGzmf6QOz95sc



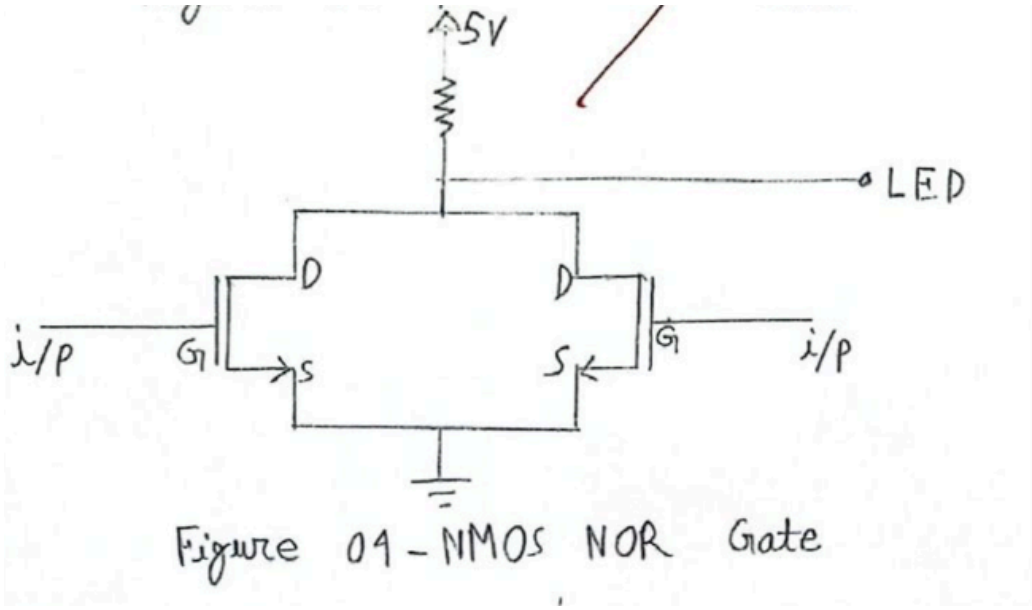
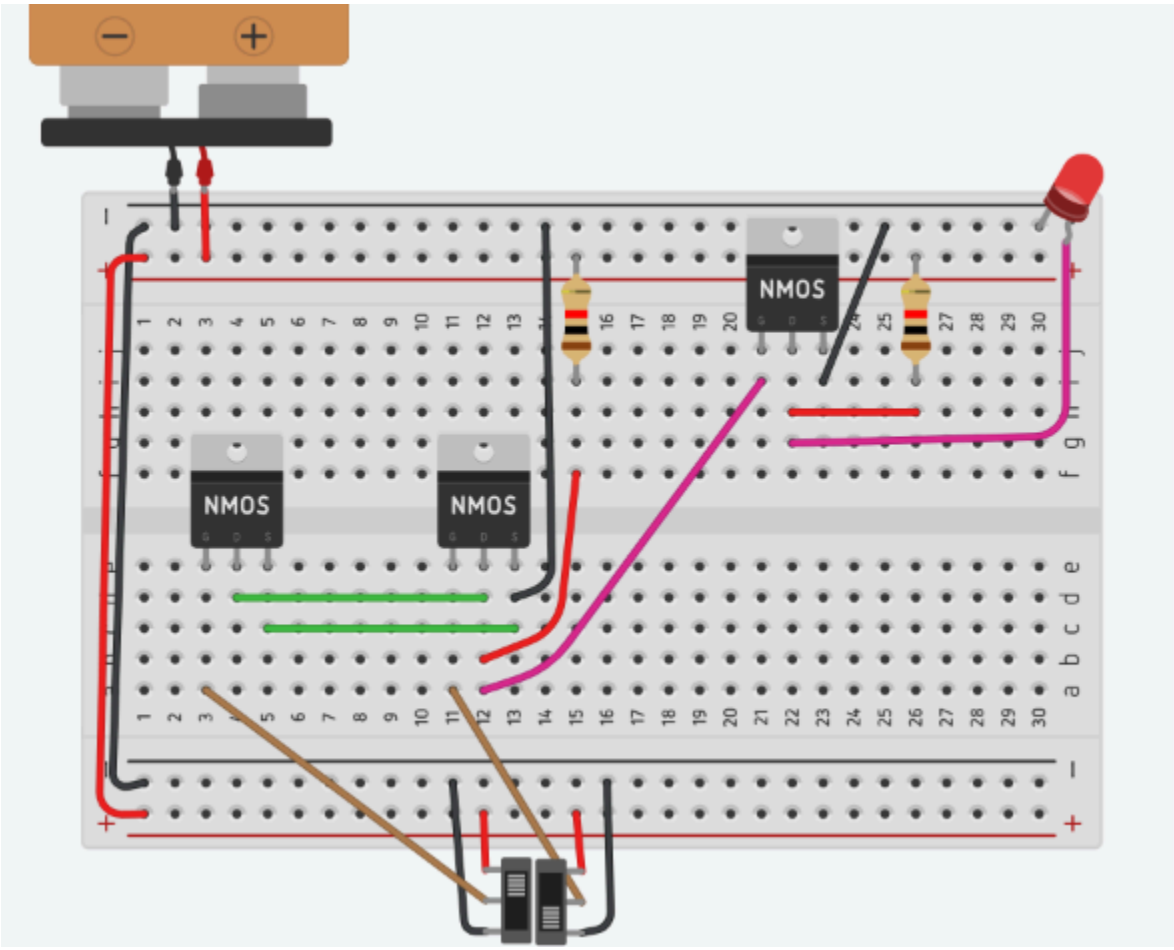


Figure 04 - NMOS NOR Gate

NMOS OR GATE

https://www.tinkercad.com/things/evEU8scOOn3/editel?returnTo=%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=vccBMF_H-BpzdRKu3CJqowapugaFHDTRdWwY5C4m70w



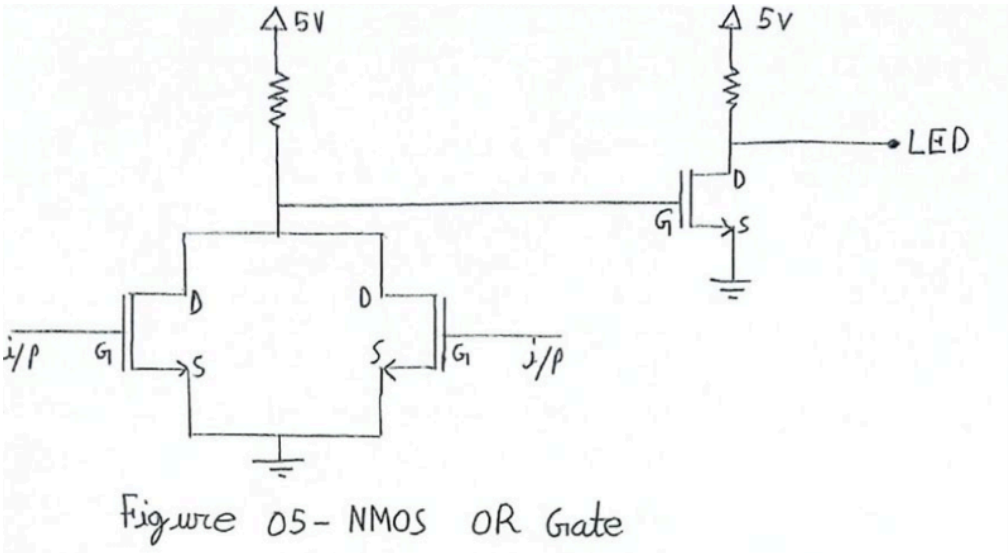
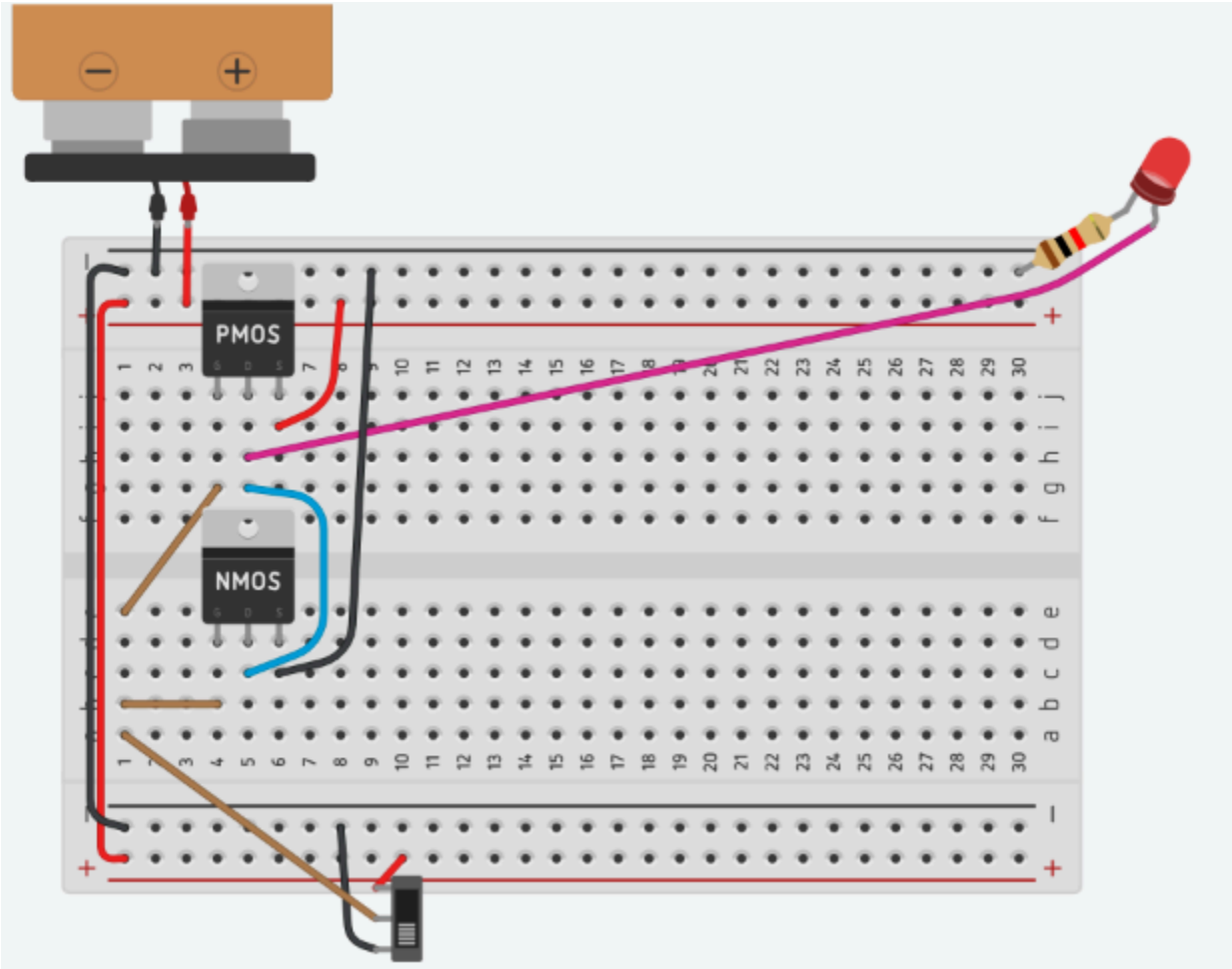


Figure 05- NMOS OR Gate

CMOS OR GATE

https://www.tinkercad.com/things/eTs8KI9I1a8/editel?returnTo=%2Fdashboard%2Fdesigns%2Fcircuits&sharecode=vRqDqxPdPOOiEbQT8BhWeENyB5BVIA4aQiRa7nMn72w



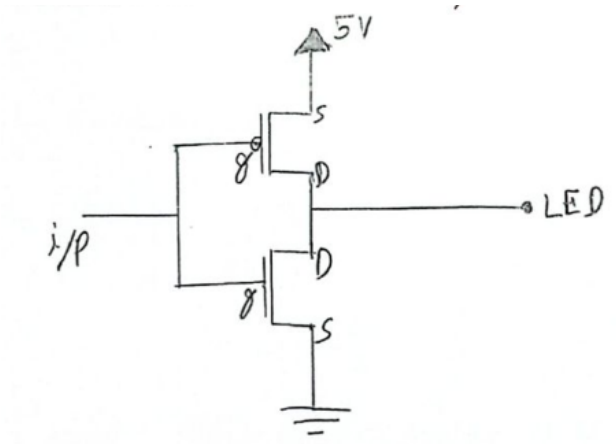
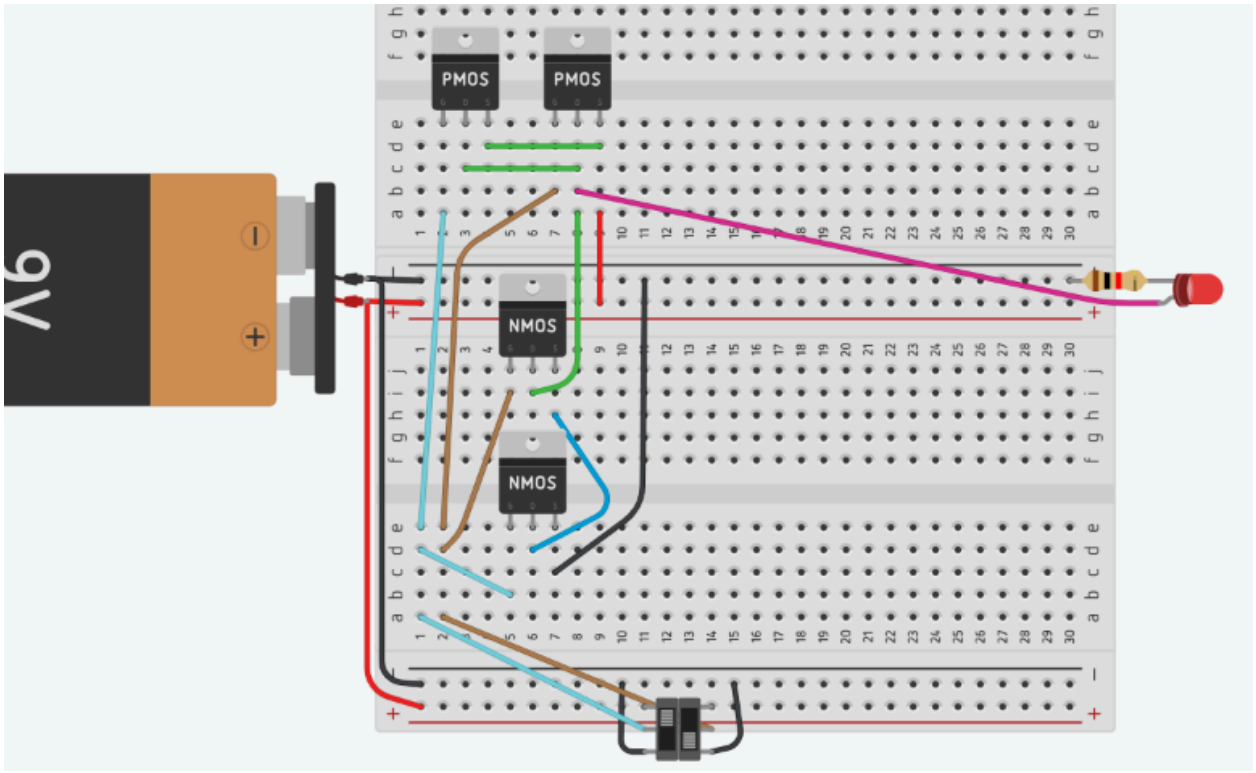
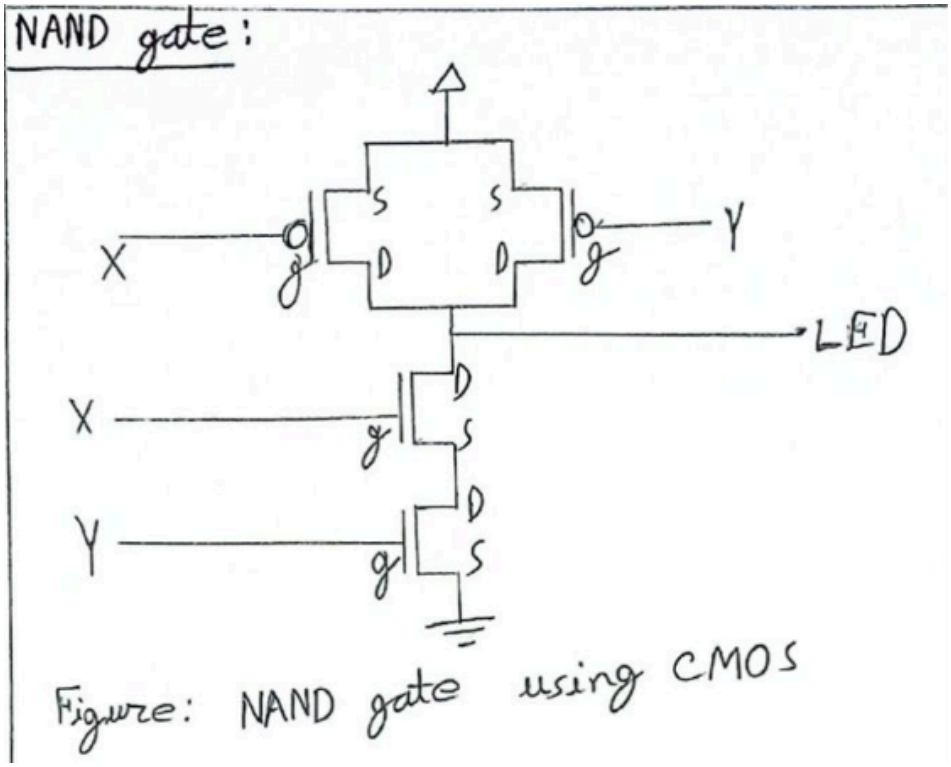


Figure: NOT gate using CMOS

CMOS NAND GATE

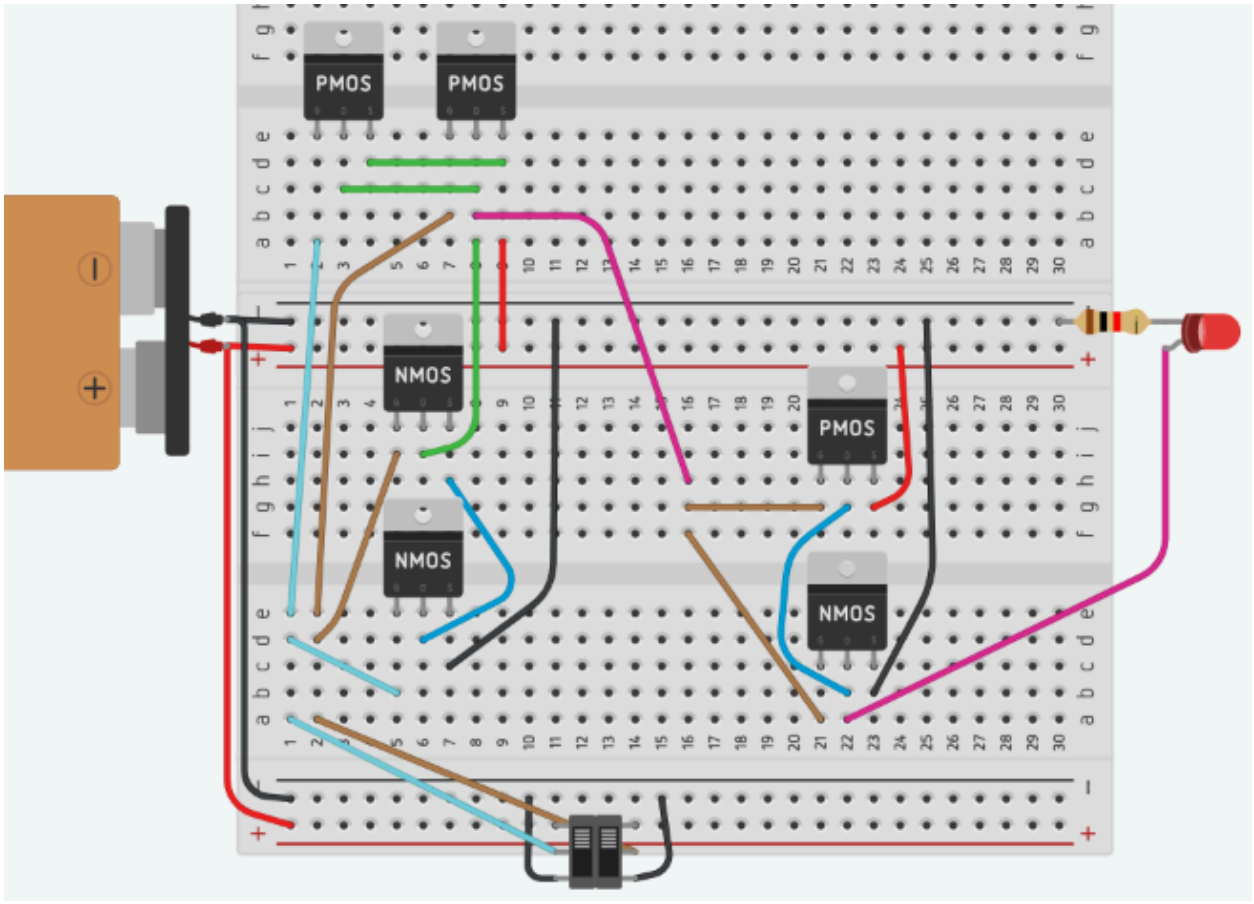
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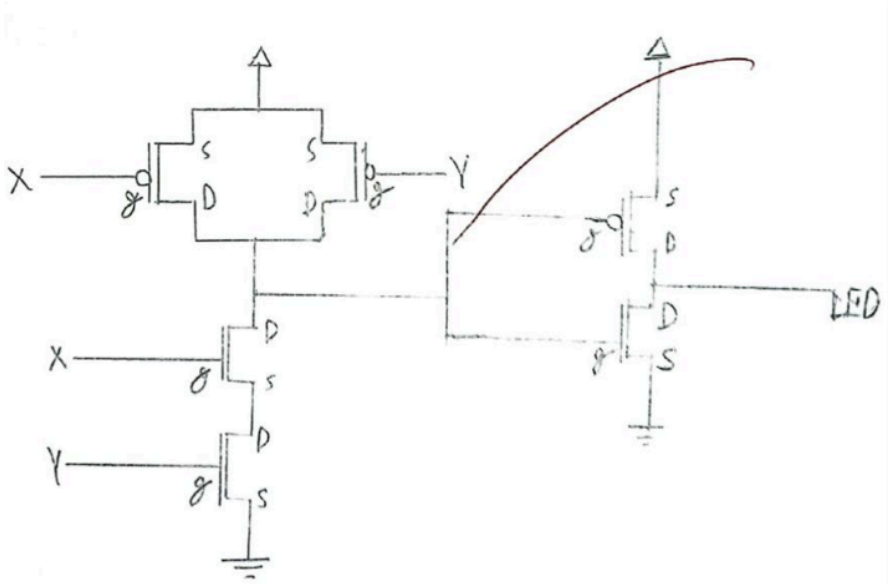




CMOS AND GATE

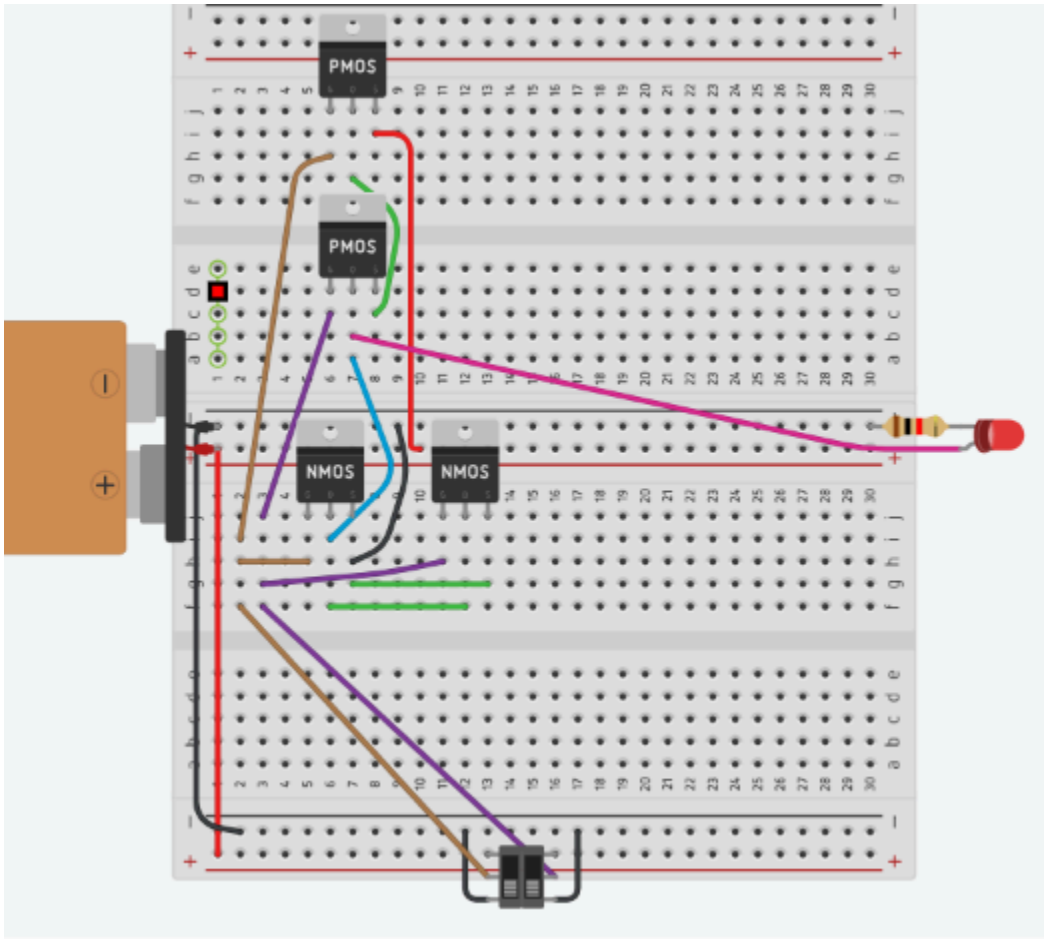
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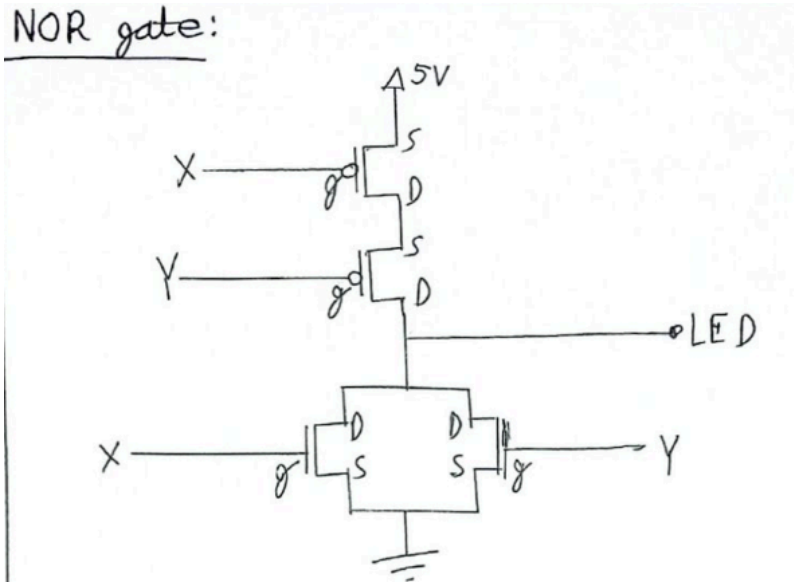




CMOS NOR GATE

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CMOS OR GATE

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